STEERING COMMITTEE STUDY

Danantara

From asset consolidator to national industrial capability platform

Performance review, DOCK-UP industrial capability framework, financing through a Natural Resource Fund and sovereign project tokenisation, construction SOE reform and multi-agentic AI

Performance | DOCK-UP | Financing | Construction SOEs and AI

October 2026

Prepared by a member of the Danantara Steering Committee for Danantara's governing bodies and management

Draft for discussion · Confidential

Important notice

This document is a study prepared by a member of the Danantara Steering Committee to support discussion among Danantara's stakeholders. It is a draft for discussion and does not represent a decision, policy or position of Danantara, its organs or the Government of Indonesia.

Basis of preparation. The study is based on six source documents (D1–D6, listed in Section 2.3) and on public references cited in those documents. We have not independently verified information provided in the source documents, except where we state that we have recalculated or reconciled figures. Where sources conflict, we report the conflict (Appendix B) rather than resolve it by assumption.

Nature of the work. This is not an audit, review or other assurance engagement; not a valuation; not a legal, tax or accounting opinion; and not an official rating. Proposed designs, thresholds, weights, timelines and financing structures are recommendations for consideration and require validation, legal review and approval by the competent organs before implementation.

Evidence classes. Figures are labelled with evidence classes A (audited), B (official reported), C (announcement, forecast or estimate), D (illustration or assumption) and NR (not rateable). Class C and D figures must not be read as realised results.

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Conflicts of interest. Three sources (D4, D6 and parts of D3) centre on maritime and defence and on PT PAL. The author's current and prior roles relevant to PT PAL and to the preparers of D4 and D6 are disclosed in the covering note to the Steering Committee, and readers should weigh the maritime and defence examples accordingly.

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Contents

01 **Executive summary** 4
02 **Introduction, scope and approach** 8
03 **Danantara today: mandate, structure and activity** 10
04 **Performance evaluation** 13
05 **What Danantara is expected to deliver** 18
06 **DOCK-UP: building national industrial capability** 22
07 **Anchor projects as the demand engine** 28
08 **Financing channel 1: the Natural Resource Fund** 32
09 **Financing channel 2: sovereign project tokenisation** 37
10 **An integrated financing architecture** 41
11 **Review of the general-contractor SOEs** 45
12 **Improving business management with multi-agentic AI** 49
13 **Governance, organisation and decision rights** 55
14 **Performance management and measurement** 61
15 **Key risks and mitigations** 64
16 **Integrated roadmap** 66
17 **Recommendations by stakeholder** 68

Appendix A: Metric dictionary 70
Appendix B: Discrepancies in the source documents 73
Appendix C: Calculations 74
Appendix D: Glossary 75
Appendix E: References 78

01
Executive summary

Danantara has the mandate, the capital and the legal foundation to drive national industrialisation. It has not yet built the mechanism that turns capital into domestic industrial capability. This study proposes that mechanism, how to finance it, and how to manage it.

1.1 The answer in one page

Danantara's visible activity to date is weighted towards **financial investment and asset consolidation**. Its mandate to drive **national industrialisation** is not yet visible in public evidence. That evidence shows institutional activation, access to international capital markets and a set of project milestones. It does not yet show risk-adjusted investment returns, attributable value creation in state-owned enterprises (SOEs), or realised industrial impact. None of these can be assessed today, because the data needed have not been made publicly available or reconciled [D1].

We recommend that Danantara reposition itself as a **National Industrial Capability Platform**. In that model:

- national strategic projects become a **demand engine**;
- the **DOCK-UP** scheme (Develop, Off-taker, Certification, Upgrading) turns that demand into qualified domestic suppliers;
- a small **Strategic Industry and Project Delivery unit** at the level of Danantara's Executive Body integrates the portfolio. This unit implements the National Industrial Transformation Office (NITO) concept;
- **three linked scorecards** measure whether capital is producing financial return, stronger enterprises and additional industrial capability [D1, D2, D3, D4].

Two complementary channels should finance the platform:

- A **Natural Resource Fund (NRF)** mobilises long-term capital from the state's share of resource wealth, at fund level.
- **Sovereign project tokenisation**, to be tested first through PT PAL's PALCapital proposal, channels capital to individual projects and suppliers against verified milestones.

Any instrument either channel issues against future state cash flows is, in substance, a liability. It must be subject to the same debt, disclosure and stress-testing discipline as other borrowing [D1, D2, D5, D6]. Both channels are complements, not the main source of funds. Even on generous assumptions they cover well under a tenth of the stated Rp2,091 trillion portfolio. Most of the funding must come from project-level debt and co-investors mobilised against Danantara's equity (Section 10.3).

Two operational reforms must run in parallel:

- The **construction SOEs** must return to a disciplined contractor model, after estimated 2025 losses of about **Rp28 trillion** (an estimate that audited figures must replace) and a fall of about 45% in the infrastructure budget [D3].
- Danantara should deploy a **multi-agentic AI project operating system**, starting with the construction SOEs. Every claimed result should then be traceable to its capital, baseline, owner and evidence [D3; D1].

Cost of inaction. Without a conversion mechanism, Danantara's strategic capital will mostly buy imported equipment and foreign technology. Its industrial mandate would then remain an announcement, and its performance would remain unmeasurable.

Rp2,091 tn
Stated value of the 40-project portfolio (class C, unreconciled; detail tables total Rp2,073 tn)

≈ Rp28 tn
Estimated 2025 losses of four construction SOEs (class C estimate)

7.7%
Average natural-resource share of state revenue, 2015–2024 (19.3% in 2004–2014)

US$150 m
Proposed PALCapital pilot, released in four milestone tranches (proposal)

1.2 Five findings

#	Finding	Evidence	Implication
1	**Market access is demonstrated; outcomes are not yet measurable.**	US$1.5 bn DIM notes at a weighted 5.65% issue yield and ~3.07× peak cover. Risk-adjusted investment returns are NR; attributable SOE value creation and realised industrial impact cannot yet be assessed [D1].	Publish an opening-balance and performance bridge before any aggregate performance claim.
2	**Implementation is weighted towards asset consolidation and financial investment.**	Advanced manufacturing and supplier development show large gaps; technology ownership is not yet proven [D3 p.1].	A deliberate capability mechanism (DOCK-UP) is required; sector coverage alone does not create capability.
3	**There is no cross-arm portfolio layer for the 40 strategic projects.**	No integrated register linking the 40 projects to leaders, decision stage and results was available to the review. The summary and detail tables differ by Rp18.93 tn and allocate differently across clusters [D2].	Reconcile one register and assign one accountable owner per project by Day 45.
4	**Law 16/2012 creates mandated domestic demand in defence; the supplier base is not ready.**	Imported defence equipment must carry trade return, local content and/or offset of at least 85%, including at least 35% local content and/or offset [D4]. D4 finds that many candidate SMEs sit at TRL 6–8 / MRL 3–5.	Use anchor projects to release volume only to suppliers with verified readiness (Section 6.9).
5	**The construction SOEs are under severe financial stress.**	Estimated 2025 losses: WIKA Rp10.14 tn, PP Rp8.0 tn, Adhi Karya Rp5.4 tn, Waskita Rp4.48 tn. Infrastructure budget Rp110.95 tn → Rp61.38 tn [D3 p.3].	Confirm with audited figures; separate legacy assets; restore cash discipline; install an AI-supported project control system.

1.3 What we recommend

Reposition

Become a National Industrial Capability Platform

Adopt a logic in which capital creates demand, demand develops suppliers, and verified upgrading creates capability. Organise activity into five portfolios with different success measures (Chapter 5).

Build

Make DOCK-UP the single scheme for all anchor projects

Use TRL and MRL as entry tests and gates G1–G4 as common rules across integrators. Release order volume only on verified readiness. Pilot with one integrator, 20–30 SMEs and 3–5 component groups (Chapters 6–7).

Finance

Run two channels under one financing discipline

NRF at fund level, based only on proven and probable reserves and the present value of the state's share. Tokenised project notes at project level, released per milestone direct to suppliers. Financing follows readiness, and mobilising co-investment is the decisive lever (Chapters 8–10).

Reform

Return construction SOEs to a disciplined contractor model

Diagnose with audited data, then separate legacy concessions into an asset vehicle. Bid selectively. Protect cash through the advance-payment-to-collection cycle (Chapter 11).

Digitise

Deploy a multi-agentic AI project operating system

Three-month MVP with six priority agents integrated read-only with existing ERP, including a bid/no-bid agent. Pilot on 2–3 projects, then scale hub-and-spoke across SOEs (Chapter 12).

Govern

Sequence: start with a unit, decide any formal position on evidence

Establish the Strategic Industry and Project Delivery unit and test it for 90 days. Review a formal Executive Body position no earlier than month 12, against agreed criteria (Chapter 13).

1.4 Decisions requested

Day 0 is the date on which the Head of the Executive Body approves the platform model in principle. All timings in this report run from Day 0 unless a calendar date is given.

(a) Decisions within Danantara's authority

Decision	Who decides	When
Approve in principle the National Industrial Capability Platform model and the Strategic Industry and Project Delivery unit, with a charter that grants no universal approval authority	Head of the Executive Body; Supervisory Board informed	Day 0
Approve the reconciled register of the 40 projects as the single denominator for all reporting (CFO interim owner until the unit is staffed)	Head of the Executive Body	Day 45
Adopt DOCK-UP gates G1–G4 and the pilot design, in coordination with KKIP, the Ministry of Defence and one integrator; pilot starts in month 4	Head of the Executive Body; integrator's board for its contracts	Day 60
Mandate a diagnostic of the four construction SOEs using audited FY2025 data	COO / DAM with the SOE boards	Start Day 30; report Day 90
Launch the three-month AI MVP (pilot live about Day 120)	COO / DAM; sponsoring SOE board	Start Day 30
Adopt the three-scorecard framework with NR treatment, 80% coverage rule and gates	Supervisory Board, on the Head's proposal	Day 90
Confirm or adjust the unit design on the basis of measured decision times and blockers	Head of the Executive Body	Day 90
Approve the PALCapital Phase 1 pilot only after all six pre-conditions are certified (Section 9.5)	DIM investment committee; PT PAL organs; within OJK's sandbox process	Condition-bound; earliest Q1 2027

(b) Proposals for the Government's and regulators' consideration

Proposal	Submitted by Danantara to	Danantara's submission date
That a legal, valuation and fiscal review of the NRF concept precede any instrument design (Chapter 8)	The Government, through the appropriate channel	Day 90
That resource-backed and receivable-backed instruments be classified by substance in fiscal-risk reporting	Ministry of Finance	Day 90
That consent procedures be defined for assigning state contract receivables to milestone escrow accounts	Ministry of Finance and Ministry of Defence	Before PALCapital approval
That DOCK-UP gates be written into framework and offset contracts for defence programmes	Ministry of Defence and KKIP	Day 60

1.5 Resources requested and what success looks like

Resource	Scale (proposed)	Basis
Unit staff	Unit head, three MDs and four cluster leaders, initially seconded	Section 13.3
DOCK-UP pilot	Assessor pool, 5–8 companion engineers, matching grants, C1–C3 co-funding and first-loss capacity for 20–30 SMEs	Section 6.9
AI MVP	14 technical staff and 6–8 business experts for three months; GPU and data servers or equivalent sovereign-cloud capacity	Section 12.6
NRF review	Independent legal, reserves-valuation and fiscal advisers	Section 8.8

Marker	By month 12	By month 24
Register and scorecards	Reconciled register; first quarterly scorecards with ≥ 80% coverage in at least one stream	Published annual portfolio report; all three streams rateable
DOCK-UP	First G1 orders placed (pilot started month 4)	≥ 60% of pilot SMEs at G1 and ≥ 25% at G2; second integrator decided
Construction SOEs	Audited diagnostic complete; legacy-asset plan approved; bid discipline in force	Positive operating cash conversion in at least two companies
AI platform	MVP evaluated; Phase 2 under way in 2–3 SOEs	All construction SOEs connected; net benefits independently verified
Financing	NRF Stage 0 complete if the Government proceeds; PALCapital pilot decision taken on the six pre-conditions	First domestic institutional investors in project notes

Markers are proposals to be confirmed after baselines are established.

02
Introduction, scope and approach

Why this study was prepared, what it covers, and how evidence is treated.

2.1 Purpose

This study is prepared by a member of the Danantara Steering Committee to give stakeholders an independent view of Danantara's current condition and performance, and to recommend how Danantara can meet the expectation that it will **raise the capability of national strategic industries** and **contribute measurably to economic indicators and social welfare**.

The study is advisory. It carries no decision authority, and its recommendations take effect only through decisions of the competent organs of Danantara and, where relevant, the Government.

2.2 Questions addressed

#	Question from the terms of reference	Where answered
1	What is Danantara's current condition and performance?	Chapters 3 and 4
2	What is Danantara expected to deliver for strategic industries, the economy and society, and how should that be measured?	Chapters 5 and 14
3	How can the DOCK-UP framework upgrade national industries, using anchor projects in minerals, infrastructure, maritime, defence and other industries?	Chapters 6 and 7
4	How should projects be funded through (a) a Natural Resource Fund and (b) sovereign project tokenisation?	Chapters 8, 9 and 10
5	What is the condition of the general-contractor SOEs, and how can business management be improved using multi-agentic AI?	Chapters 11 and 12
–	What organisation, governance, risk controls and roadmap are needed to implement the above?	Chapters 13, 15, 16 and 17

2.3 Source documents

The study relies on six source documents provided for this review. They are cited in square brackets throughout. Each source cites its own underlying references; where we rely on one of those, we cite it as, for example, [D1/S05].

Table 1: Source register

Code	Document	Date	Nature and use
D1	Danantara performance study: strategic and profitable investment, operating efficiency and growth, and additional economic development	6 Oct 2026	Public-evidence assessment with 45 definitions (39 result metrics and 6 governance gates); basis for Chapters 4 and 14

Code	Document	Date	Nature and use
D2	Evaluasi Tata Kelola Danantara dan Organisasi Industri Strategis: percepatan 40 proyek strategis dan opsi Wakil Kepala Badan Pelaksana	2 Oct 2026	Governance and organisation review; 40-project data checks; RASCI; basis for Chapter 13
D3	Danantara Platform Kapabilitas Industri Nasional (8 pages)	7 Oct 2026	Diagnosis, platform model, NITO, construction SOE condition and multi-agentic AI design; basis for Chapters 5, 7, 11 and 12
D4	DOCK-UP: Membangun Pemasok UKM Siap Produksi untuk Industri Maritim dan Pertahanan Indonesia	1 Oct 2026	Supplier-development framework using TRL and MRL; basis for Chapter 6
D5	Indonesia Natural Resource Fund: Executive Summary	25 Sep 2025	NRF concept, targets and eight strategic steps; basis for Chapter 8
D6	PALCapital: Sovereign Maritime Tokenization Engine, Rev B	Undated (Phase 1 targeted Q4 2026)	Tokenised MoD-contract receivable for PT PAL; basis for Chapter 9

2.4 Evidence discipline

Every number in this report carries, explicitly or by context, one of five evidence classes adopted from D1. An announcement is never read as a result, and a missing value is never treated as zero or as failure.

Class	Meaning	Permitted use
A	Audited or independently assured	Highest confidence, provided scope, period and definition match
B	Official reported observation	Usable with an explicit assurance limitation; includes company releases and calculations derived from them
C	Announcement, forecast, commitment or estimate	Evidence of intent or of a milestone; not evidence of realised profit, output or impact
D	Analyst illustration or assumption	Method design and sensitivity analysis only; excluded from performance scoring
NR	Not rateable	Missing, incompatible or insufficiently assured data, or no approved benchmark; different from zero and from failure

2.5 Limitations

- This is an analytical study for discussion. It is **not an audit, a valuation opinion, a legal opinion or an official rating**, and it does not approve any organisation, appointment or financing structure.
- Public evidence was reviewed as of early October 2026. The Danantara reports page displayed “Coming Soon”; this does not prove that no financial statements exist [D1/S01].
- The 40-project portfolio is a working scope taken from discussion material. It is not a validated register and is not equated with the legally designated list of National Strategic Projects [D2].
- Where source documents contradict each other or themselves, we say so and do not resolve the conflict by assumption. Appendix B lists every discrepancy found.
- Proposed thresholds, weights, timelines and organisational designs are **recommendations**, not current Danantara policy.

03
Danantara today: mandate, structure and activity

What Danantara is set up to do, how it is organised, and what it has visibly done so far.

3.1 Legal foundation

Danantara, formally the Badan Pengelola Investasi (BPI) Daya Anagata Nusantara, was established through the amendment of the SOE Law (Law 19/2003) by Law 1/2025. The SOE Law was amended again by Law 16/2025. BPI operates under Government Regulation (PP) 10/2025, as amended by PP 19/2026. Its two organs are the Supervisory Board (Dewan Pengawas) and the Executive Body (Badan Pelaksana), which is led by the Head (Kepala) [D2/R1–R3; Law 1/2025 is public law not cited in D2, to be verified].

In this report, “BPI level” means the agency level above the three operating arms. The regulations:

- separate the Supervisory Board from the Executive Body;
- place Executive Body decisions with the Head;
- provide for members of the Executive Body to be appointed by the President;
- differentiate the purposes of the investment holdings.

Provision	What it establishes	Relevance to this study
PP 10/2025 Art. 12–18	Organs, membership and decision-making of the Executive Body; Head takes Executive Body decisions (Art. 18); members appointed by the President (Art. 13); risk portfolio assigned to a member (Art. 14)	Any new senior role must fit this structure; a title does not by itself transfer decision rights
PP 19/2026 Art. 21A, 21B, 29A–29C, 32A	Planning, holding purposes and transitional provisions; Art. 32A starts certain plan submissions from financial year 2028	Does not defer all governance obligations to 2028; does not explicitly create a Deputy Head position
Law 16/2025	Amends Arts. 3X, 3Y, 3AA; deletes Art. 9G	Business judgement protection (Art. 3Y) is conditional, not immunity; arguments relying on Art. 9G must be updated

Legal summaries are taken from D2 and are not a legal opinion. Article references should be verified by Danantara Legal before reliance.

3.2 Organisation

The Danantara ecosystem comprises BPI, led by its Executive Body, and three operating arms. The COO, CIO and CTO of BPI also head the three arms, which creates dual capacities that must be managed through separate decision documents and conflict rules [D2/R4–R7].

Unit	Mandate	Primary performance question
BPI (led by the Executive Body)	Strategy, portfolio, risk, finance, technology; Head takes Executive Body decisions	Is the ecosystem as a whole creating value above its cost of capital and delivering its development mandate?

Unit	Mandate	Primary performance question
DAM: Danantara Asset Management (led by the COO)	Ownership and transformation of the operating SOE portfolio	Are SOEs becoming more productive and generating economic profit?
DIM: Danantara Investment Management (led by the CIO)	Commercial investment, capital allocation and investment partnerships; issued the US$1.5 bn notes	Are investments earning net returns above an appropriate benchmark?
DDMF: Danantara Development Management Fund (led by the CTO)	Project preparation, structuring, capital mobilisation and delivery for national development and public services	Are development projects economically justified and delivered, with public support quantified?

BPI's published structure already includes a CTO and Managing Directors for Industrialization, Technology Advancement and Strategic Technology Initiatives [D2/R4]. The capability gap is therefore not the absence of named functions. It is the absence of a portfolio layer that integrates them across arms around specific projects and suppliers.

3.3 2026 investment plan and priorities

Danantara's 2026 plan targets investment of **up to US$14 billion**, split roughly **50% public markets** and **50% strategic projects in the real sector**. Stated priorities are renewable energy, digital infrastructure, health and food security. The broader strategic sectors named are mineral downstreaming, renewable energy, chemicals, data centres, food security, waste-to-energy and agriculture [D3 p.1].

A 50/50 split is reasonable for liquidity and diversification. But public-market investment, on its own, does not create new production capacity, Indonesian technology, domestic machines and components, or significant industrial value added [D3 p.1]. The strategic half of the plan is where the industrialisation mandate will succeed or fail.

3.4 Visible activity to date

Table 2: What public evidence shows, and what it does not yet show

Evidence	Class	What it shows	Test still outstanding
US$1.5 bn DIM international notes, June 2026: two US$750 m tranches at 5.35% and 5.95%; issuer-reported spreads 32 and 34 bps; peak orders ~US$4.6 bn [D1/S05]	B	Market access at a principal-weighted 5.65% yield, ~33 bps average spread and ~3.07× cover	All-in cost after fees and hedging; use of proceeds; returns on the assets funded
Danantara and INA conditional subscription of US$200 m in an US$800 m CA–EDC chemicals project, March 2026 [D1/S06]	C	A capital-commitment stage equal to 25% of project value; operations targeted 2027	Conditions satisfied; individual capital split; cash paid; import substitution actually achieved
PSEL Bogor Raya 1 waste-to-energy: construction inauguration and PPA, September 2026; Rp2.8 tn; 1,500 t/day [D1/S07]	C	A substantive development milestone; operations targeted mid-2028	Certified physical progress, full financing, commissioning, sustained output
Six Phase-I downstream groundbreakings, total announced value up to US$7 bn (MIND ID, Feb 2026) [D1/S08]	C	Programme activation	Stage, financing share and benefits per project; not US$7 bn disbursed by Danantara
Eight Batch-II waste-to-energy conditional awards, July 2026 [D1/S30]	C	Pipeline progression	Eight conditional awards are not eight operating plants

Evidence	Class	What it shows	Test still outstanding
Governance and accounting review announced March 2026 [D1/S04]	C	Reform action consistent with an asset-quality review	Verified operating gains, completed remediation, durable controls
Statement of 2 July 2026 that SOEs completed FY2025 statements and consolidation continues; official statement of 12 May 2026 on consolidation and audit [D1/S02; D2/R8]	B	Reporting process under way	Audited consolidated statements and a portfolio return series

Reading announcements correctly

A signed power-purchase agreement does not establish financial close. A construction inauguration does not establish earned progress. A transfer of asset ownership does not establish appreciation. Higher dividend collection may reflect payout decisions or legacy earnings rather than productivity [D1].

3.5 Milestones in the sources

Date	Milestone	Source
1 Dec 2025	2026 work plan and budget published	D1/S03
6 Feb 2026	Six Phase-I downstream groundbreakings (MIND ID), announced value up to US$7 bn	D1/S08
3–5 Mar 2026	CA–EDC investment agreement with INA; governance and accounting review announced	D1/S04, S06
12 May 2026	Official statement on consolidation and audit	D2/R8
12 Jun 2026	US$1.5 bn DIM international notes	D1/S05
2 Jul 2026	SOEs complete FY2025 statements; consolidation continues	D1/S02
21 Jul 2026	Eight Batch-II waste-to-energy conditional awards	D1/S30
17 Sep 2026	Construction of PSEL Bogor Raya 1 begins; PPA signed	D1/S07
Early Oct 2026	Reports page on the official website still shows “Coming Soon”	D1/S01

3.6 How peers disclose performance

Mature sovereign investors publish return series that let stakeholders judge performance. Danantara has not yet done so. The comparison below shows reporting practice only. It is not a return target, because mandates, currencies and portfolios differ [D1/S17–S19].

Institution	What is published	Example figures
Temasek (Singapore)	Annual review; total shareholder return over 5, 10 and 20 years; risk-adjusted cost-of-capital comparison	To 31 Mar 2026, in SGD: 5-year 4.6%, 10-year 7.1%, 20-year 6.8%
Khazanah (Malaysia)	Annual review with annual and multi-year returns, reported alongside strategic activity	FY2025: 5.2% for the year; 7-year annualised 6.1%
Danantara	Announcements of financing, commitments and milestones; reports page “Coming Soon”	No portfolio return series published

04
Performance evaluation

Danantara's performance assessed across its three mandates (investment, SOE operations and industrialisation), together with governance and execution readiness.

4.1 Overall verdict

Stream	Rating	Current assessment	What would move the rating up
Investment	Access demonstrated; returns NR	Financing and commitments evidenced; portfolio return performance **not rateable (NR)**	Audited net-asset bridge; return series by capital pool against approved benchmarks
SOE operations	Mixed; contribution unverified	Selected outcomes are **mixed**; Danantara's incremental contribution unverified	Economic-profit and real unit-cost data against a without-intervention baseline
Industrialisation	Not yet evidenced	Project delivery has advanced; most capacity and impact figures remain **prospective**	Commissioned capacity, verified domestic value added and capability milestones
Governance	Foundations in place; effectiveness unverified	**Foundations available; effectiveness of implementation unverified** [D2]	Documented decision rights, independent risk and audit lines, conflict registers, measured decision times

Rating scale used in this report: Demonstrated → Partly evidenced → Not yet evidenced → NR (not rateable). Ratings are qualitative judgements on public evidence.

The strongest supported conclusion is progress in **mobilising and organising execution**. The largest unresolved question is how much **incremental value** those actions have produced after financing costs, risk and public support [D1].

4.2 Investment performance

Funding access is measurable; investment success is not yet. The June 2026 notes show that DIM can raise US-dollar funding at spreads close to the sovereign curve. That is a financing result. It says nothing about coupon cost after hedging, about the weighted average cost of capital, or about whether the assets financed earn more than that cost. Borrowing below an equity hurdle does not make an investment value-creating [D1].

Table 3: Four capital pools that must be measured separately

Pool	Primary measure	Boundary rule
Inherited operating SOEs	ROIC / economic profit; for banks, ROE with capital adequacy	Value at transfer and effective control date; legacy liabilities separated
Liquid financial investments	Net time-weighted return versus policy benchmark; drawdown and liquidity	Market return, FX and external cash flows reconciled separately

Pool	Primary measure	Boundary rule
Private and strategic investments	Net IRR, DPI/TVPI and public-market equivalent; project financial NPV	Tracked by vintage; residual value separated from realised cash
Development-supported investments	Financial sustainability plus ENPV/EIRR and additional outcomes	Subsidies, guarantees and concessionality identified explicitly

Judgement. Financing capability: demonstrated for the observed issue. Strategic deal origination: evidenced by selected commitments. Risk-adjusted portfolio profitability: NR. The required next evidence is an audited opening-to-closing net-asset bridge and a reconciled cash-flow and valuation series for each pool [D1].

4.3 SOE operating performance

A purposive sample of large SOEs shows both improvement and deterioration. Periods, ownership shares and accounting scopes differ, so these figures must not be summed into Danantara earnings [D1].

Table 4: Selected SOE observations (class B)

Entity / metric	Comparable observation	Reading
BRI group net income	FY2024 restated Rp60.306 tn → FY2025 Rp57.132 tn (−5.26%)	Profit contracted on the published comparable base [D1/S10]
BRI bank-only CIR / gross NPL	CIR 37.87% → 38.92%; NPL 2.94% → 3.29%	Efficiency and credit quality weakened; drivers need analysis
Mandiri parent-attributable profit	FY2024 Rp55.78 tn → FY2025 Rp56.30 tn (≈ +0.93%)	Modest growth; does not by itself prove economic-profit improvement [D1/S11]
Telkom FY2025	Revenue Rp146.7 tn; EBITDA Rp72.2 tn; reported net income Rp17.8 tn; normalised net income Rp22.7 tn; Telkom reports a 9.5% profit decline on its revised basis	Earnings require the restatement and normalisation bridge [D1/S12]
PLN (state electricity company) H1 revenue	H1 2025 Rp281.89 tn → H1 2026 Rp327.15 tn (+16.05%)	Growth must be split into volume, tariff, compensation and other effects [D1/S13]

The valid test of SOE transformation is **incremental economic profit**: normalised NOPAT less a capital charge, compared with a credible without-intervention path, paired with real unit costs, service reliability and cash flow. For banks, the test is ROE relative to cost of equity, credit quality and capital headroom. Fewer legal entities, higher dividend extraction or headcount reduction should not be rewarded unless productivity and service outcomes are verified [D1].

4.4 Industrialisation performance

Project evidence is promising but almost entirely pre-operational. The sector-by-sector picture shows where Danantara is active and where the industrialisation mandate is not yet being served.

Table 5: Evaluation by priority sector

Sector	Danantara's strategic direction	Visible activity	Assessment
Financial returns	Sustainable return, diversification, public and private markets	Public-market investment and global exploration	**Active**
SOE optimisation	Improve performance of state assets	Portfolio management, dividends, restructuring	**More mature**

Sector	Danantara's strategic direction	Visible activity	Assessment
Industrial downstreaming	Nickel, aluminium, bauxite, copper and other value chains	Smelters, alumina, caustic soda, downstream projects	**Strategic, but limited to resource processing**
Energy transition	Renewable energy and waste-to-energy	WtE and clean-energy initiatives	**Strategic; project economics and technology ownership unclear**
Digital industry	Data centres and digital infrastructure	Data-centre platforms with global operators	**Infrastructure-oriented, not yet technology-industrial**
Food security	Agriculture, fertiliser, poultry, agribusiness	Integrated agriculture and livestock projects	**Strategic, not always technology-intensive**
Advanced manufacturing	Machinery, electronics, shipyards, aerospace (policy priority)	Public evidence still limited	**Large gap**
Industrial supplier development	Not yet a clear core KPI	Public evidence of supplier localisation and domestic engineering limited	**Large gap**
Technology ownership	Innovation and competitiveness	Foreign partners still dominant in several projects	**Not proven**

Source: D3 p.1. Assessments are qualitative and based on public evidence.

Four tests of genuine industrial development

Test 1
Additionality
Would the project have happened at the same scale, timing, quality and financing terms without Danantara? Name the specific financing, coordination or technology constraint that was removed.

Test 2
Competitiveness
Can the plant produce at acceptable cost, quality and reliability after support tapers? Measure unit economics in contestable markets, not protected sales.

Test 3
Capability
Can domestic teams operate, improve and reproduce critical processes? Require verified design rights, process control, supplier qualification and repeat commercial sales.

Test 4
Distribution and durability
Are jobs sustained, suppliers more productive, services affordable and environmental gains verified over the life of the asset?

National GDP growth (5.11% in 2025 against 5.03% in 2024) cannot be attributed to Danantara without a counterfactual and evidence of incremental output. A public-investment announcement is not an estimate of GDP added [D1/S14].

4.5 Governance foundations

Assessed against the OECD SOE Guidelines 2024, Danantara's design provides an adequate base for good governance. Its effectiveness depends on internal instruments and their application, which public evidence cannot verify [D2].

Dimension	Evidence and assessment	Action needed
Clarity of organs	PP separates the Supervisory Board and the Executive Body. Supervisory foundation available.	Check the board manual, reserved matters, committee charters and appointment instruments
Owner and regulator mandates	BP BUMN and BPI have different legal roles; rights over each company follow ownership and articles.	Document when BPI acts as owner, coordinator, investor or implementer of an assignment
Commercial versus development	PP 19 differentiates holding purposes; DDMF publishes a development mandate.	Apply financial targets, impact targets and sources of support appropriate to each mandate
Industrial execution competence	CTO and MDs for Industrialization, Technology Advancement and Strategic Technology Initiatives exist.	Test engineering, contracting, commissioning and supply-chain integration capacity before adding positions
Risk at organ level	Art. 14 requires a risk portfolio among Executive Body members; an MD Risk title does not prove compliance.	Verify the Presidential Decree, portfolio assignment and the risk function's access and committee position
Conflicts of interest	PP requires disclosure and exclusion of conflicted parties.	Keep a conflict register, recusal minutes and an alternative decision path when the leadership is conflicted

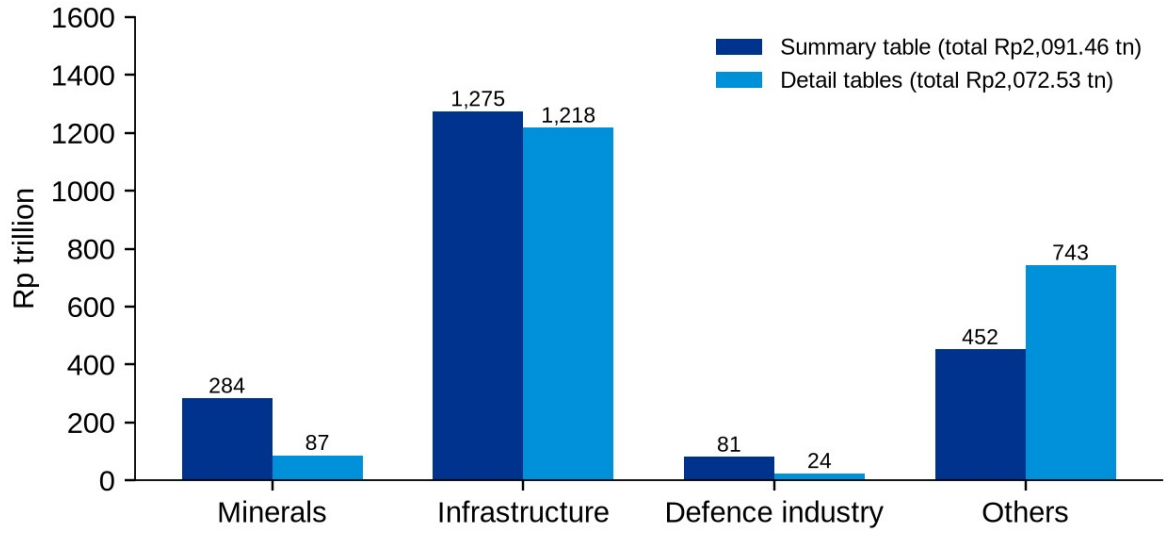
4.6 Readiness to execute the 40 strategic projects

Readiness to execute the 40 projects cannot be demonstrated on six dimensions. No integrated register linking each project to a leader, a decision stage and its results was available to the review [D2]. In our judgement this is the most important gap to close first.

Dimension	Assessment on available evidence	Requirement
Project accountability	No integrated register linking the 40 projects to leaders, decision stage and results was available to the review	One accountable outcome owner per project; one register of decisions and blockers
Capital authority	Head decides for the Executive Body; holding and company decisions are separate layers	Authorisation map by entity, object and stage; signing authority separated from business approval
Guarantees and fiscal support	A development mandate does not automatically confer a state guarantee or capital injection (PMN)	Show the guarantor, its capacity to pay, the legal basis for support and the contingent cost
Procurement and partners	Procurement policy and partner selection must be examined at implementing entities	Test beneficial ownership, related-party transactions, competition, sole-source rationale and change control
Transparency and audit	Consolidation and audit under way (statement of 12 May 2026); current audit opinions not assessed	Review audit reports and follow-up; report portfolio and impact consistently
Measuring success	Investment lists and job targets do not prove benefits	Measure operations, utilisation, cost, net impact, technology mastered and post-investment results

The 40-project data do not reconcile

Figure 1: The 40-project portfolio: summary table versus detail tables



Source: D2 §1, based on the 40-project infographic (U1). Detail tables converted from Rp million. Evidence class C.

Cluster	Summary (Rp tn)	Share	Detail (Rp tn)	Share	Difference (Rp tn)
Minerals	283.85	13.6%	86.87	4.2%	−196.98
Infrastructure	1,274.57	60.9%	1,217.92	58.8%	−56.65
Defence industry	81.10	3.9%	24.36	1.2%	−56.74
Others	451.94	21.6%	743.38	35.9%	+291.44
Total	**2,091.46**	**100%**	**2,072.53**	**100%**	**−18.93**

The totals differ by **Rp18.93 trillion**, and the cluster allocations differ far more: “Others” is Rp291 trillion larger in the detail tables, while minerals is Rp197 trillion smaller. Labour figures in the infographic are estimates, not outcomes. Units and scope must be confirmed with the data owner before any figure is used for budget approval [D2]. Until then, completion, mobilisation and impact rates have no valid denominator.

4.7 Root causes

All four strands of the source work point to the same four root causes [D3 p.2; D1; D2; D4]:

Root cause 1

Capital without a conversion mechanism

Capital is available, but there is no clear mechanism to turn investment into national industrial capability.

Root cause 2

Import-heavy projects with little transfer

Many projects are import- and construction-led, with limited technology transfer and supplier development. Examples include PT PAL's reported 75% imported materials [D6] and candidate SMEs at MRL 3–5 [D4].

Root cause 3

SOEs working in silos

SOEs often act alone and are not integrated into an industrial platform [D3 p.2]. No cross-arm portfolio layer or integrated register was available to the review [D2].

Root cause 4

Impact not measured

Effects on GDP, technology and exports are not measured with a defensible method. Announcements stand in for outcomes [D1].

05
What Danantara is expected to deliver

Danantara is expected to strengthen national strategic industries and to contribute measurably to economic indicators and social welfare. This chapter turns that expectation into a target state that can be managed and measured.

5.1 From expectation to measurable outcomes

The expectation placed on Danantara has three parts. Each must be defined as an **additional** outcome that would not have happened without Danantara. Each needs its own measure, because a single profit or dividend number cannot represent all three [D1].

Expectation	What success looks like	Primary measures (Chapter 14; definitions in Appendix A)
Boost national strategic industry capability	Domestic firms can design, make, certify, maintain and improve critical products and processes at competitive cost, and repeat sales without permanent protection	Suppliers passing DOCK-UP gates; verified technology milestones (D06); domestic value-added share (D05); commercial capacity utilisation (D07)
Contribute to economic metrics	Additional real value added, net foreign-exchange contribution, productivity growth and private capital mobilised, net of displacement and public support	Additional domestic value added (D02); net FX contribution (D04); supplier productivity (D09); attributable private mobilisation (I10); economic return above the social hurdle (D01)
Social impact	Sustained quality jobs, SMEs integrated into strategic supply chains, reliable and affordable services, and verified environmental gains	Sustained additional employment (D03); reliable affordable service delivered (D11); net lifecycle GHG reduction (D10); public support per additional outcome (D12)

Order of magnitude: what localisation could contribute

No baseline exists yet for these indicators. To show the order of magnitude at stake, the table below estimates what shifting part of the anchor portfolio's imported goods to verified domestic suppliers could contribute. It is a **class D illustration**: every assumption is stated and must be replaced with project data.

Assumption (illustrative)	Value
Goods and equipment as share of anchor capex	45%
Imported share of those goods today	60% (PT PAL reports 75% for its materials [D6])
Domestic value added per rupiah of domestic purchase	40%
Import content of domestic supply	30%
Value added per full-time job-year	Rp250 million
Exchange rate	Rp16,500 per US$

Table 6: Localisation scenarios (class D)

Share of imported goods shifted to DOCK-UP suppliers	10%	20%	30%
Domestic purchases per Rp100 tn of capex (Rp tn)	2.7	5.4	8.1
Additional domestic value added per Rp100 tn (Rp tn)	1.1	2.2	3.2
Whole portfolio (Rp2,091 tn): domestic purchases (Rp tn)	56	113	169
Whole portfolio: additional value added (Rp tn)	23	45	68
Whole portfolio: net foreign exchange saved (Rp tn / US$ bn)	40 / 2.4	79 / 4.8	119 / 7.2
Whole portfolio: job-years supported	≈ 90,000	≈ 181,000	≈ 271,000

Class D illustration. Whole-portfolio figures accrue over the build period, not per year. They scale the per-Rp100 tn results by 20.9 and use the unreconciled portfolio total.

Even the modest scenario is likely to exceed the cost of running DOCK-UP by a wide margin, to be confirmed once the pilot is costed. The figures are gross of displacement and must not be reported as GDP added (Section 14.4).

5.2 Strategic objective and the five values

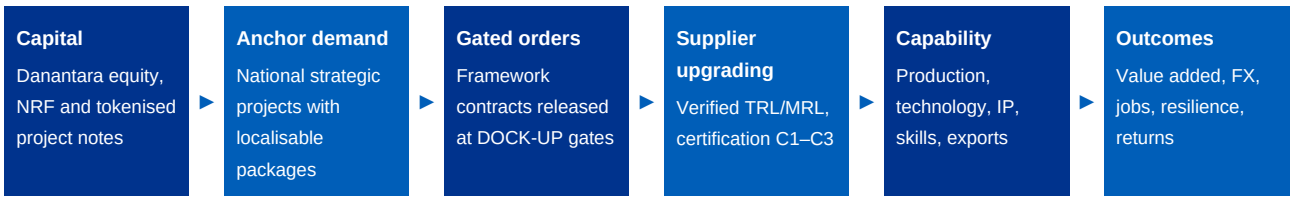
Proposed strategic objective

Generate sustainable returns while building national industrial capability that raises Indonesia's productivity, technology, resilience and competitiveness [D3 p.2].

Value	What Danantara contributes	Illustrative indicator
Economic value	Returns above the cost of capital; additional value added; private capital mobilised	Net IRR above hurdle; ENPV; mobilisation ratio
Industrial value	New domestic production capacity and qualified suppliers	Qualified suppliers at G3/G4; utilisation of new capacity
Technology value	Technology mastered and IP owned or controlled in Indonesia	Independently accepted capability milestones; enforceable operating and improvement rights
Strategic value	Resilience of critical supply chains, especially defence and energy	Share of critical components with a qualified domestic source; through-life support capacity
Social value	Quality jobs, SME participation, services, environment	Sustained FTE; SMEs in supply chains; service units; tCO2e reduced

5.3 Logic of change

The platform converts capital into capability through a sequence in which each step must be verified before the next is funded. This is the core of the recommended model.



The transformation is from **asset manager and investor** to **national industrial capability platform based on strategic projects and domestic demand**. Danantara contributes capital, governance and mandate; DOCK-UP converts that into production, technology, domestic suppliers, human capital and exports [D3 p.2].

5.4 Portfolio architecture

We recommend organising Danantara's activity into five portfolios, each with a different success measure. This prevents the most common error in mixed-mandate funds: judging a supplier-development programme by the yardstick of a listed-equity portfolio, or the reverse [D3 p.2; D1].

Table 7: Five portfolios with distinct measures and owners

Portfolio	Scope	Primary success measure	Proposed lead
Wealth preservation	Public equities, bonds and mature global assets	Net time-weighted return versus policy benchmark; liquidity and drawdown	DIM
Strategic infrastructure	Energy, ports, logistics, industrial estates, data centres	Financial NPV and ENPV; availability and service reliability	DDMF with DIM
Industrial platform	Anchor projects: Giant Sea Wall, 3 Million Homes, maritime, defence (D3); we propose adding mineral downstreaming	Localisation achieved through DOCK-UP; verified domestic value added; on-time, on-budget commissioning	Strategic Industry unit with project sponsors
Innovation	Semiconductors, AI, advanced materials, R&D and future technologies	Stage-gated uncertainty reduction; independently accepted technology milestones	CTO
Supplier development	MSMEs and Tier 1–3 suppliers of equipment, components and fabrication	Suppliers passing G1–G4; supplier productivity; cost parity with imports	DOCK-UP office with integrators and banks

One map from anchor project to portfolio

Readers meet several groupings in this report: anchor projects (Chapter 7), the four clusters in D2, and the five portfolios above. The table below maps them onto each other. The industrial-platform portfolio's financial returns remain the responsibility of the investing entity (DIM, DDMF or the sponsoring SOE). The unit is accountable for integration and capability outcomes, not for capital approval.

Anchor project	Cluster (D2)	Portfolio	Lead for capability	Accountable for returns
Mineral downstreaming	Minerals and materials	Industrial platform	Cluster leader, minerals	Sponsoring SOE / DIM
Giant Sea Wall	Energy and industrial infrastructure	Strategic infrastructure + Industrial platform	Cluster leader, infrastructure	Project company / DDMF
3 Million Homes	Energy and industrial infrastructure	Industrial platform	Cluster leader, infrastructure	Programme owner / sponsoring SOE

Anchor project	Cluster (D2)	Portfolio	Lead for capability	Accountable for returns
Maritime (shipyards, MRO)	Defence and maritime	Industrial platform	Cluster leader, defence and maritime	PT PAL / DEFEND ID / DIM
Defence programmes	Defence and maritime	Industrial platform	Cluster leader, defence and maritime	Integrator / Ministry of Defence contract
Chemicals, WtE, data centres, food	Other strategic manufacturing	Strategic infrastructure or Industrial platform	Sector leads once split	DIM / DDMF / sponsoring SOE
MSME and Tier 1–3 suppliers	All clusters	Supplier development	DOCK-UP office in the unit	Banks and guarantors for credit risk

Whether the Giant Sea Wall and the 3 Million Homes programme are in the 40-project register must be confirmed during reconciliation.

5.5 Cross-sector technology enablers

Six enabling technologies named in D3 p.2 cut across the anchor projects (descriptions ours). They should be prioritised in the innovation portfolio because they raise capability in several clusters at once:

- **Power electronics and semiconductor design**: drives, converters and control boards used in ships, energy systems and defence electronics.
- **Industrial AI and automation**: process control, quality inspection and the project operating system described in Chapter 12.
- **Advanced materials (steel, composites)**: marine-grade steel, armour and composite structures.
- **Renewable energy and energy storage**: battery materials and storage linked to mineral downstreaming.
- **Machine-tool industry**: the capital goods that let SMEs move from MRL 5 to MRL 8.
- **Digital industry and software**: engineering software, PLM, supplier portals and cybersecurity.

5.6 Design principles

1. **Demand before subsidy.** Support follows a real order book from an anchor project, not the reverse.
2. **Readiness before volume.** Orders, finance and certificates are released by verified readiness, never by quota [D4].
3. **One register, one denominator.** Every project, phase, supplier and financing instrument has one identifier in one register [D1, D2].
4. **Finance follows the gate.** Each financing instrument is matched to the gate a project or supplier has reached.
5. **Substance over label.** An instrument that must be repaid is a liability, whatever it is called and wherever it is booked [D1, D2].
6. **Protection is temporary.** Preference for domestic suppliers must be temporary [D4]; we propose a defined exit.
7. **Measure the additional, not the gross.** Report Danantara's estimated contribution with ranges, not gross project totals [D1].

06
DOCK-UP: building national industrial capability

A gated framework that turns guaranteed domestic demand into qualified, production-ready suppliers, measured with Technology Readiness Levels (TRL) and Manufacturing Readiness Levels (MRL).

6.1 The problem: demand exists in law, the supplier base does not

85% / 35%
Minimum trade-return and local content / offset for imported defence equipment (Law 16/2012 Art. 43), rising 10 points every 5 years

64.2 million
MSMEs in Indonesia: ~62% of GDP and ~97% of employment (KemenkopUKM, 2021), yet few are in strategic supply chains

25–30 years
Service life of a warship; maintenance, spares and upgrades often cost more than construction

Source: D4 p.2.

D4 describes a vicious circle that can only be broken by a concrete programme in which development, off-take, certification and upgrading are linked by gates that combine TRL and MRL [D4]. In our reading, the circle works like this. Integrators will not order from SMEs that are not qualified. SMEs cannot invest in qualification without orders. Banks will not lend without either.

6.2 A common language: TRL and MRL

TRL answers “does it work?”. MRL answers “can it be made every time?”. For SMEs, the recurring problem is scale-up (MRL) of technology they already master; large industry is needed to help with proven methods. Large industry is also needed to mature technology (TRL) [D4].

TRL	Technology readiness	MRL	Manufacturing readiness
1–2	Principles observed; nothing built	1–3	Feasibility discussed; no process
3	Proof of concept: laboratory experiment succeeds	4	Laboratory scale; hand-made or basic tools
4–5	Validated in a controlled lab, then a simulated environment	5–6	Pilot process using production-representative methods
6–7	Prototype demonstrated in a relevant environment; actual testing	7	Low rate; repeatable quality at low volume
8–9	Proven, qualified and in routine use	8	Pre-production: tooling, supply chain and quality control proven
		9–10	Full sustained production at competitive cost

6.3 Diagnosis: four supplier profiles

Each SME is mapped on both scales to determine the next form of help. The four profiles are archetypes, not a complete partition of every TRL/MRL combination [D4].

TRL 1–4 · MRL 1–3

Explorer

The idea still has to be proven. Right support: research grants, BRIN and university laboratories.

TRL 6–8 · MRL 3–5

Prototype-rich, production-weak

Most SMEs are stuck here. Right support: anchor orders, shared equipment and companion engineers. This is the core of DOCK-UP.

TRL 4–5 · MRL 6–8

Production-ready, shallow technology

Reliable production; own product not yet validated. Right support: R&D partners and product certification.

TRL 8–9 · MRL 8–10

Qualified supplier

Both proven. Right support: long-term framework contracts and exports.

6.4 The four elements of DOCK-UP

Element	What it does	Exit or milestone
D: Develop	Baseline TRL/MRL; support targeted at the weakest MRL aspect. At platform level: identify industrial opportunities, design projects with domestic content, prepare pre-FEED and FEED [D3 p.2].	Exit at G1 when MRL 6 is reached
O: Off-taker	3–5 year framework contracts; volume released in stages at G1, G2 and G3. At platform level: aggregate demand from government, SOEs and anchor projects to give volume certainty to local suppliers.	Tranche 1 pilot order (G1) → low-rate supply (G2) → serial supply and spares (G3)
C: Certification	C1 quality = MRL 7 (ISO 9001); C2 class = MRL 8 (user approval, ISO 3834/9606 welding, traceability); C3 defence-industry quality plus security. At platform level: technical standards, TKDN based on capability, testing and validation of domestic products.	C1 at G2; C2 at G3; C3 for defence-grade, sensitive items
UP: Upgrading	Progression from build-to-print → build-to-spec → design-and-build. At G4 the SME's own TRL counts. At platform level: technology transfer, innovation and efficiency, exports and IP.	G4: own product TRL 7+, MRL 9, adopted on a ship class

The governing rule

Readiness, not promises or commitments, unlocks the next step. An integrator may not order more than the SME has been proven able to deliver. Scores are never self-declared: independent assessors verify, the integrator reviews the evidence, and scores are reassessed every year [D4].

6.5 Integrated gate model

Each gate sends the same message to the SME, the buyer and the bank. That shared signal is what allows non-government finance to enter.

Table 8: DOCK-UP gates

Gate	Readiness	Evidence	Unlocks	Main support
Entry	Baseline TRL/MRL	Self-assessment verified by an independent assessor	Development plan and mentoring	Companion engineers, training
G1	MRL 6	First-article inspection passed	Tranche 1: pilot order	Matching grant, shared facilities

Gate	Readiness	Evidence	Unlocks	Main support
G2	MRL 7 + C1	Small repetitive batch; ISO 9001	Tranche 2: low-rate supply	Supply-chain finance, KUR
G3	MRL 8 + C2	User approval; ISO 3834/9606; traceability	Tranche 3: serial supply and spares	Investment credit, first-loss guarantee
G4	Own product TRL 7+; MRL 9	Product adopted on one ship class	Build-to-spec, design-and-build, exports	Corporate venture, R&D super-deduction, LPEI

Source: D4 p.6.

6.6 Two routes through DOCK-UP

The instruments are the same; the order differs. A build-to-print SME follows the **MRL route**; an SME with its own product follows the **TRL route**. The two illustrative SMEs below are taken from D4 and are not real companies.

Stage	MRL route: “Sidoarjo Steel Manufacturing” (bearings and valves, from household goods to frigates)	TRL route: “Sensor Karya Mandiri” (infrared sensors, from handheld cameras to warships and missiles)
Baseline	Civil MRL 7, warship MRL 4. Mass production for bicycles and household goods; weak on materials and testing.	Civil foothold: TRL 8–9 handheld thermal cameras generating revenue.
Build capability	Months 1–6: companion engineer; certified marine-grade material, qualified machining, hydrostatic testing.	Defence requalification, TRL 5 → 7: shock, vibration, salt spray, EMC; IRST sea testing.
First gate	Month 7, **G1 (MRL 6)**: first warship valve passes inspection; pilot order of 20 valves.	Defence-grade production, MRL 5 → 8: calibration, traceability, C3, export control.
Scale	Months 8–14, **G2 (MRL 7 + C1)**: on-time batches; valves and bearings for one ship set.	IRST on a warship, TRL 8, **G4**: sea trials on a warship.
Mature	Months 15–24, **G3 (MRL 8 + C2)**: standardised for one frigate class.	Heat-seeking missile sensor: sub-supplier to the DEFEND ID missile programme.

6.7 Economics: why verified readiness brings in private finance

Lower cost of capital

A multi-year off-take plus a verified MRL score gives banks a basis for credit assessment.

Economic lot sizes

Standard components across several ship types and DEFEND ID companies allow pooled volume.

FX and multiplier effects

Local content stays in the country and cannot be withheld by another state.

Through-life value

Law 16/2012 mandates domestic MRO; Tranche 3 includes spares, which over a 25–30 year life can exceed the build value.

Learning-curve illustration

D4 illustrates an **85% learning curve**: each doubling of cumulative volume lowers unit cost to 85% of its previous level. Pooling orders across DEFEND ID companies moves SMEs down the curve faster. The table shows what that implies (class D illustration).

Cumulative volume (× first lot)	1×	2×	4×	8×	16×
Unit cost (% of first lot)	100	85.0	72.3	61.4	52.2
If first-lot cost is 140% of import price	140	119	101	86	73

On these assumptions, an SME starting 40% above the import price reaches parity after about 2.1 doublings, roughly **4× the first-lot volume**. This is why order aggregation across integrators matters more than any single subsidy, and why protection should be **temporary**: once parity is reached, the SME must sustain MRL 8 on efficiency, and the integrator can open DOCK-UP to the next cohort [D4].

Finance matched to gates

D4 links finance instruments to gates: matching grants, KUR (people's business credit), supply-chain finance, first-loss guarantees, the R&D super-deduction under PP 45/2019, and LPEI export finance. Chapter 10 integrates these into the wider financing architecture.

6.8 Technology and assessment

Technology priority	Content
Process quality first	Qualified welding, heat treatment, navy-standard coating and non-destructive testing (NDT)
Tiered technology transfer	Offset agreements name the packages that are sub-licensed to DOCK-UP suppliers
Digital integration	Supplier portal connected to the integrator's ERP: drawings, revisions, traceability and gate records
Information security	Controlled access and cybersecurity, included in C3 (defence grade) according to sensitivity



6.9 Way forward and the single biggest risk

1. Establish DOCK-UP as the standard model for SME participation, coordinated by KKIP (Defence Industry Policy Committee).
2. Make TRL (TKT) and MRL assessment the entry screen.
3. Adopt gates G1–G4 as common rules across integrators.
4. Pilot with one integrator, 20–30 SMEs and 3–5 component groups.
5. Write DOCK-UP packages, by gate, into framework contracts and offset agreements.
6. Co-fund C1–C3 certification; tie KUR, supply-chain finance and LPEI to gate achievement.
7. Build an independent assessor pool and a shared readiness register.

Who runs DOCK-UP

KKIP sets defence-industry policy. Under the unit, a cross-sector **DOCK-UP office** maintains the gate rules, the assessor pool and the shared readiness register. Integrators and anchor-project sponsors place orders and manage supplier performance. Banks and guarantors take credit decisions on the basis of verified gates.

Gates and statutory minimums

Law 16/2012 sets quantitative minimums for trade return, local content and offset. Readiness gates do not replace those minimums. They decide **which** suppliers receive volume. Where verified domestic readiness falls short of the legal minimum, the gap should be closed through offset and technology-transfer packages that feed DOCK-UP suppliers, not by orders to suppliers who are not ready. Compliance with the minimums continues to be certified by the competent authority.

Pilot budget structure and exit targets (proposed)

Cost element	Driver for a 20–30 SME pilot	Funding source
Independent assessors	Pool of about 10–15 accredited assessors; entry assessment plus annual reassessment per SME	Danantara / KKIP programme budget
Companion engineers	About 1 engineer per 4 SMEs (5–8 engineers) for 12–24 months	Integrator and Danantara, shared
Matching grants and shared facilities	First-article tooling and test equipment for SMEs approaching G1	Danantara grant window; NRF supplier window once open
Certification co-funding	C1 (ISO 9001) for all SMEs reaching G2; C2 and C3 for those approaching G3	Integrator, Danantara and SME, shared
First-loss guarantees	A first-loss tranche on bank credit to SMEs at G2–G3	Guarantee provider and banks; NRF supplier window once open

Exit target (to be confirmed), months from pilot start	Month 12	Month 18	Month 24
Pilot SMEs at G1 or above	≥ 60%	≥ 75%	≥ 80%
Pilot SMEs at G2 or above	–	≥ 25%	≥ 40%
First-article or batch rejection rate	Baseline set	Below agreed limit	Below agreed limit
Bank credit extended on gate evidence	First facilities	Growing	Without grant support for G2+

Costs are not estimated here because unit rates for assessors, engineers and certification must come from the integrator and KKIP. The table gives the cost drivers for budgeting.

Single biggest risk

Treating DOCK-UP as a **quota** rather than a gated path of technology and manufacturing readiness would destroy the model. Orders placed before verified readiness can cause quality failures on warships and damage the credibility of the whole programme. Integrators must mitigate volume and quality risk [D4].

6.10 Extending DOCK-UP to every anchor project

DOCK-UP was designed for maritime and defence. D3 proposes **one DOCK-UP scheme for all anchor projects**. The gates transfer directly; what changes by sector is the certifying body, the qualification evidence and the off-taker.

Sector	Typical DOCK-UP packages	C1/C2 evidence (illustrative)	Off-taker
Maritime and defence	Valves, bearings, piping, cable harnesses, sensors, coatings, MRO	ISO 9001; class approval (BKI); welding ISO 3834/9606; C3 security	PT PAL and DEFEND ID integrators; MoD programmes
Coastal infrastructure (Giant Sea Wall)	Precast concrete, steel piles, sheet piles, gates, pumps, dredging support	Product standards and material test certificates; owner's engineer approval	Project companies and EPC contractors
Housing (3 Million Homes)	Modular and prefabricated components, HVAC, electrical systems, fittings and pumps. Mature bulk materials such as cement are, in our assumption, lower DOCK-UP priorities.	Product standards; factory production control for modular units	Housing programme developers and construction SOEs
Minerals and downstreaming	Process equipment parts, refractories, wear parts, electrical, maintenance services	Owner qualification; plant-specific vendor approval	Smelter and refinery operators
Energy and WtE	Boiler and grate parts, electrical balance of plant, control panels, O&M services	Utility vendor qualification	Project companies; utility

Certification evidence for sectors other than maritime and defence is our illustrative proposal and must be confirmed with the relevant authorities and owners.

07
Anchor projects as the demand engine

National strategic projects in minerals, infrastructure, maritime, defence and other industries are used as guaranteed demand to build domestic supply chains.

7.1 The anchor-project principle

An anchor project is a large, long-duration programme with predictable procurement in which Danantara or its SOEs participate. Its value to industrial policy is not only the asset it builds but the **order book it creates**. Each anchor project should publish a localisation plan identifying which packages can be supplied domestically, at which gate, and when.

Table 9: Anchor projects and the industrial demand they create

Anchor project	Industrial platform	Demand created for domestic industry
Giant Sea Wall	Maritime and coastal industry platform	Concrete and steel structures; shipyards and marine equipment; pumps, water gates and local materials; flood protection and coastal resilience
3 Million Homes programme	National housing industry platform	Cement, steel, glass and ceramics; modular housing components; plumbing, electrical and HVAC systems; environmentally friendly materials
Maritime industry	Shipyards, components and MRO	Commercial and special-purpose vessels; engines, propulsion and marine systems; MRO and docking; ship components, electronics and equipment
Defence industry	Defence equipment and support systems	Land, sea and air platforms; electronics, sensors and radar; engines, propulsion and special materials; system integration and maintenance
Mineral downstreaming	Minerals and materials platform	Nickel, aluminium, bauxite and copper processing: smelters, alumina, caustic soda; process equipment, maintenance and logistics services
Other strategic industries	Chemicals, energy, digital, food	CA–EDC chemicals, waste-to-energy, data centres, agriculture and fertiliser; to be split into proper clusters once the register is complete

Source: D3 pp.1–2; D1/S06–S08; D2 §9.

7.2 Four initial clusters

D2 proposes four initial clusters aligned with the portfolio material. “Others” must be split once a complete register exists, because food, digital, health and other projects need different expertise. Civil shipbuilding and defence should be organised as separate work packages even within one cluster [D2].

Cluster	Anchor projects (examples)	Portfolio value, summary table (Rp tn)	Cluster leader's first tasks
Minerals and materials	Nickel, bauxite–alumina–aluminium, copper, battery materials	283.85	Map raw materials, markets, technology and shared facilities; set capability tests
Energy and industrial infrastructure	Giant Sea Wall, energy, WtE, ports, industrial estates, 3 Million Homes	1,274.57 (infrastructure)	Package localisable items; align construction SOEs with DOCK-UP
Defence and maritime	Submarines, frigates, LPDs, hospital ships, MRO, DEFEND ID programmes	81.10	Launch the DOCK-UP pilot; link offsets to packages
Other strategic manufacturing	Chemicals (CA–EDC), digital, food, health	451.94 (others)	Split by sector; assign specialist leads

Values are from the unreconciled summary table and are shown for orientation only (see Section 4.6).

7.3 Capability tests by cluster

Each cluster needs two sets of tests: financial and operating tests, and industrial-capability tests. A pilot or demonstration must never be mistaken for industrial success [D1].

Cluster	Financial and operating tests	Industrial capability and outcome tests
Battery materials, recycling, storage	Yield, recovery, delivered cost, qualification costs, working capital, offtake, IRR, scale-up capex	Battery-grade qualified output, verified local value added, operating IP rights, repeat sales, lifecycle emissions
Alumina, aluminium, minerals	Ore and energy costs, recovery, utilisation, logistics, downside metal-price returns	Incremental domestic processing value, net FX after imported inputs, supplier capability, energy and carbon intensity
CA–EDC and chemicals	Power and feedstock intensity, utilisation, product spreads, safety, cash generation	Commercial import displacement, export competitiveness, reliable domestic supply
Waste-to-energy and clean infrastructure	Whole-life service cost, contracted revenue, debt service, residual waste liabilities	Waste actually processed, net lifecycle GHG, local air quality, affordability, recycling effects
Maritime, shipbuilding, logistics	Order-book quality, on-time delivery, lifecycle cost, repair turnaround, warranty liabilities	Certified design and manufacturing capability, domestic component value, export orders, logistics productivity
Advanced technology, strategic manufacturing	Stage-gated R&D capital, qualified demand, cost-down path, commercial-scale readiness	Production quality, domestic engineering autonomy, competitive repeat orders

Source: D1, "Apply the framework to strategic industries".

7.4 Linking project gates and supplier gates

Two gate systems must interlock. Project gates (D2's G0–G6, here labelled **P0–P6** to avoid confusion) govern whether a project proceeds. Supplier gates (D4's **G1–G4**) govern whether a supplier receives volume. The **localisation plan** is the document that connects them.

Project gate	Evidence to proceed (D2)	Localisation requirement (proposed)
P0 Mandate validation	Project identity, mandate basis, sponsor, scope, cost data, decision path	Flag whether the project is an anchor with localisable packages
P1 Priority and options	Financial case and national-impact case; alternatives including not building; links between projects	Long list of localisable packages with current import share
P2 Investment readiness	Studies, technical, land, permits, IP, E&S, market, offtake, risk, cost, sources of capital	Localisation plan: packages by PBS, candidate suppliers with verified TRL/MRL baselines, required-on-site dates
P3 Capital decision	Committee recommendation, dissent, conflicts, reserved approvals, conditions	Framework contracts and offset packages written into the EPC/supply contracts, with tranches tied to G1–G3
P4 Financial close and construction	Binding funding, conditions precedent, effective contracts, permits, baseline schedule, change control	Tranches released only as suppliers pass gates; fall-back import route pre-approved for schedule protection
P5 Commissioning and operation	Performance tests, safety, asset acceptance, operator, supply-chain readiness, certification	Supplier performance and spares readiness verified; through-life support contracts
P6 Results evaluation	Final cost, actual benefits, capacity, technology, emissions, corrective action or exit	Verified domestic value added and capability milestones reported against the plan

Contents of a localisation plan

- Package list structured by the project's product breakdown structure (PBS), so each package has a cost code, quantity and required-on-site date.
- Current import share and target domestic share per package, by supplier gate.
- Candidate suppliers with independently verified TRL/MRL baseline and development plan.
- Off-take terms: framework contract, volume tranches, pricing formula and price-parity path.
- Certification route (C1–C3) and who pays for it.
- Financing instrument matched to each gate (Chapter 10).
- Schedule protection: a pre-approved fall-back source if a supplier misses a gate, so that localisation never delays the anchor project's critical path.
- Indicators: domestic value-added share (D05), suppliers per gate, supplier productivity (D09).

7.5 Worked example: the maritime anchor

PT PAL's sovereign backlog, comprising two Scorpène Evolved submarines (US$2.16 bn), two Merah Putih frigates (AH-140, US$1.2 bn), LPDs and hospital ships, illustrates how the pieces fit. About **75% of materials are imported**, and each programme faces a working-capital gap of **US$150–300 m** because materials must be paid before milestone payments arrive [D6].



This is the integration point of the whole study. The **demand** is the defence programme; **DOCK-UP** qualifies the suppliers; **project tokenisation** finances the working capital and pays suppliers directly from escrow on

verified milestones; and the **AI project operating system** provides the evidence trail. Chapter 9 assesses the financing instrument in detail.

7.6 Worked examples for other anchors

The same steps apply to other anchors. The examples below are **illustrative (class D)**. They show the packages a localisation plan would examine first, chosen because they are commonly imported. Actual import shares must come from each project's bill of materials.

Step	Nickel or alumina smelter	Giant Sea Wall	3 Million Homes
Localisable packages to examine first	Wear parts, refractories, conveyors, pumps and valves, electrical panels, maintenance services	Pumps and water gates, steel sheet piles, mechanical and electrical works for pump stations, dredging support	Prefabricated and modular units, HVAC, electrical and plumbing fittings
Typical supplier starting point	Fabricators with general industrial MRL; process-grade quality not yet proven	Steel fabricators and pump assemblers; large-diameter pumps and gates largely imported	Many domestic producers; the gap is consistent quality at volume
Gate emphasis	MRL route: material traceability and wear-life testing	TRL and MRL routes: hydraulic performance testing for pumps and gates	MRL route: factory production control and logistics
Off-taker	Smelter operator under a 3–5 year framework contract	Project company or EPC contractor	Developers and construction SOEs through aggregated programme demand
Key indicator	Domestic share of maintenance spend; unplanned downtime from local parts	Domestic share of pump-station value; through-life O&M by local firms	Unit cost and defect rate of modular components

08
Financing channel 1: the Natural Resource Fund

A fund-level channel that converts the state's share of natural-resource wealth into long-term capital. It is assessed here for rationale, design, valuation, legal basis, fiscal substance and governance.

8.1 The proposal

The NRF executive summary positions the fund as a strategic fiscal instrument that implements Article 33 of the 1945 Constitution ("natural resources for the greatest prosperity of the people"). Through eight strategic steps it targets **direct fiscal benefits of Rp100 trillion in the first year and Rp1,000 trillion within three years** [D5].

Design element	As proposed in D5
Position	Directly under the President
Scope	Consolidate all strategic minerals and coal, with coal as the quick win; land is also listed as a field
Minimum valuation	Rp10,000 trillion (coal reserve capitalisation)
Underlying asset	Strategic mineral reserves 100% owned by the state, verified to international reporting standards (CRIRSCO: KCMI)
Mechanism	Asset securitisation, transforming natural resources into security-based assets; securitisation capped at **50% of future cash flow**
Monetisation options	(1) Sovereign asset-backed securities; (2) mineral-based sovereign notes; (3) royalty-backed financing; (4) Mineral Reserve Trust
Flows with the state budget	(1) Debt reprofiling with the APBN; (2) tax revenue to the APBN
Uses	Energy transition, economic diversification, future fund; project implementation through SPVs or SOEs
Management	Corporate management
Preparation	Targeted for completion within three months of approval in principle

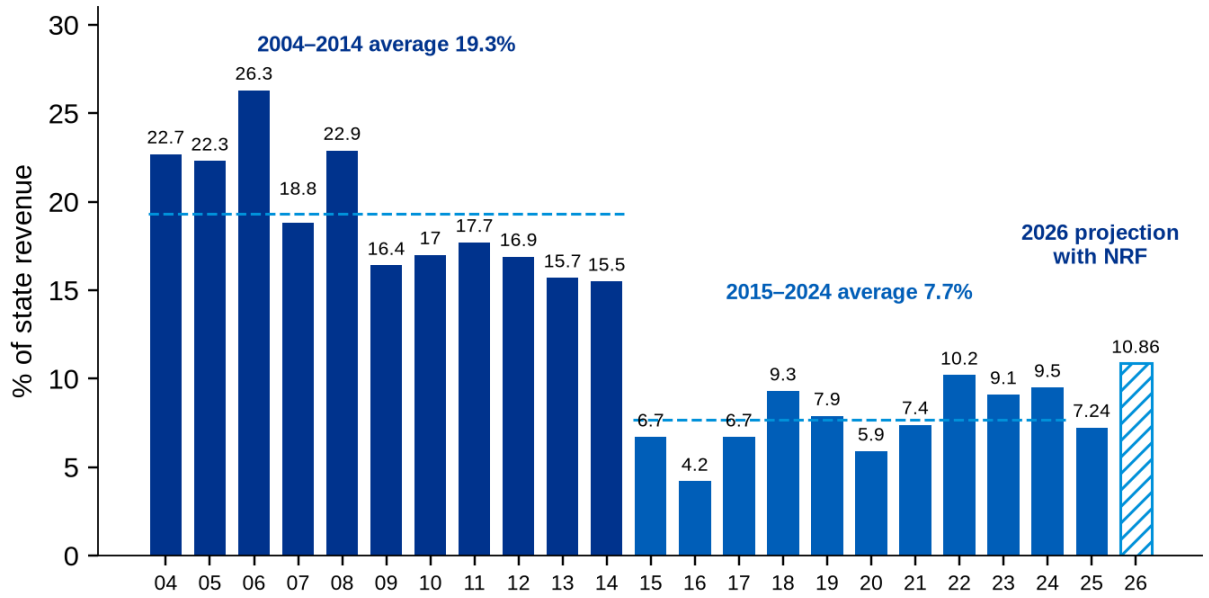
Eight strategic steps (0–2 years)

#	Step
1	Enforcement against, and nationalisation of, illegal mining
2	Review of mining business licences (IUP) and land-use rights (HGU)
3	Strengthening of the legal foundation
4	Establishment of the NRF under the President
5	Institutional restructuring of related SOEs

#	Step
6	Integration of mining systems and governance under the NRF
7	Management and use of natural-resource fund proceeds
8	Involvement of Koperasi Merah Putih (village cooperatives) and distribution of benefits

8.2 Rationale: low fiscal capture from natural resources

Figure 2: Natural-resource revenue as a share of state revenue, 2004–2026



Source: D5, values read from the chart. D5 does not state the source of the series: treat as class C until reconciled to budget data. 2026 is a projection with the NRF. The chart labels 2026 as 10.86%; the D5 text states 10.68%.

19.3% → 7.7%
Natural-resource contribution to the state budget, 2004–2014 versus 2015–2024

< 50%
2024 natural-resource revenue relative to interest payments on state debt

Rp1,684.7 tn
Market capitalisation of coal-mining companies cited in D5 (a stock; state royalty of ~Rp126.4 tn in 2024 is a flow)

< 2%
D5's figure for the state's share of total coal potential (Rp10,000 tn); method not stated

Source: D5. The 2026 projection is Rp336 tn of natural-resource benefits, equal to 10.68% of state revenue, implying state revenue of about Rp3,146 tn (our calculation).

The direction of the diagnosis is supported. The averages in D5 reconcile with its own annual data: the eleven values for 2004–2014 average 19.3%, and the ten values for 2015–2024 average 7.7%.

Two cautions apply. First, the fall in the share reflects several causes together: commodity prices, oil and gas production, changes in the fiscal regime, and growth in other revenue. It is not evidence of low capture alone. Second, D5 compares market capitalisation, a stock, with annual royalty, a flow. That is the same stock-versus-flow mismatch flagged in Issue 2 below. A sound measure of capture is state take (royalty, PNBP and tax) divided by sector economic rent or EBITDA, by commodity and year. With that decomposition, a mechanism that raises capture, saves part of the proceeds and invests part in domestic capability would be consistent with Article 33 and with international practice for resource funds.

8.3 Assessment: what must be resolved before design

The NRF concept is directionally right. Its numbers, legal basis and fiscal treatment are not yet sound enough to support a decision. Eight issues must be resolved.

Table 10: NRF issues and required resolution

#	Issue	Why it matters	Required resolution
1	**Valuation basis.** Rp10,000 tn is a potential or in-situ figure; the methodology is not stated.	Financeable value is the present value of the state's share of cash flows from proven and probable reserves, typically a small fraction of in-situ value (Section 8.5).	Independent valuation under KCMI/CRIRSCO; publish the method and the haircuts.
2	**Benefit targets.** Rp100 tn in year 1 and Rp1,000 tn in three years are described in the text as “direct fiscal benefits”. In the chart they appear under “coal capitalisation value with NRF”, next to “Coal Reserve Capitalization” of Rp10,000 tn.	Capitalisation (a stock) and fiscal benefit (a flow) are different things; they cannot be compared with budget revenue.	Restate targets as cash flows to the budget, net of forgone future revenue and costs.
3	**Legal rights.** State control under Article 33 does not by itself give BPI or an NRF the right to pledge unmined reserves or non-tax revenue (PNBP) [D2 §18].	Investors and courts will test the chain of rights; a defect voids the security.	Legal review of who holds mining rights, the right to sell and to pledge, tax and PNBP treatment, and the authorisation needed.
4	**Fiscal substance.** Securitising up to 50% of future cash flow borrows against future revenue.	An instrument with an obligation to repay or a fixed coupon is a financing liability; an off-budget label does not remove fiscal or contingent risk [D1, D2].	Record as debt in the fiscal-risk statement; show the present value of revenue given up.
5	**Debt reprofiling with the APBN.** Using resource-backed securities to reprofile state debt swaps one liability for another.	There is a gain only if the all-in cost is lower and the risk transfer is real.	Like-for-like comparison against SBN of the same tenor and currency.
6	**Institutional position.** An NRF directly under the President alongside Danantara risks overlapping mandates.	D1 cites the Santiago Principles for investment and risk governance [D1/S16]. The point on clear owner, governing-body and manager roles is our reading of the principles.	Define the NRF as owner/fund with a mandate letter. Professional managers, which could include Danantara arms, manage assets under contract, subject to the Government's decision.
7	**Due process.** Steps 1 and 2 (nationalising illegal mining; reviewing IUP and HGU) affect property and licence rights.	Investor confidence across the resource sector depends on legal certainty.	Enforcement and licence reviews follow transparent due process, with compensation rules and an independent appeal route.
8	**Distribution channel.** Benefit distribution through Koperasi Merah Putih mixes a programme channel with an investment vehicle.	Commingling weakens both investment discipline and programme accountability.	Route distributions through the budget or a separate programme window with its own audit.

8.4 The four monetisation options compared

Option	Strength	Weakness	Assessment
Sovereign asset-backed securities	Familiar to institutional investors; ring-fenced cash flows	Requires true sale or clear security over revenue; complex SPV and rating work	**Viable** for producing assets with stable royalty flows
Mineral-based sovereign note	Uses existing state-securities framework; simplest for domestic investors	Is plainly state debt; adds little unless revenue-linked	**Preferred first instrument** if linked to a ring-fenced revenue account
Royalty-backed financing	Directly monetises the state's existing take; measurable	Brings forward revenue that future budgets would otherwise receive	**Viable** with a cap on advance rate and a coverage test
Mineral Reserve Trust	Can hold reserves for the long term and separate ownership from operation	Hardest legally; valuation of unmined reserves is contested	**Later stage**, after rights and valuation are settled

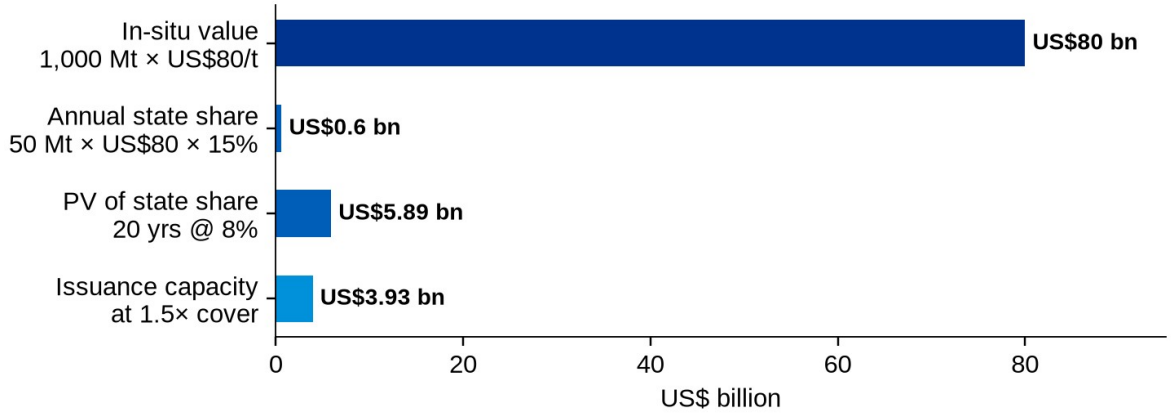
Assessment is our analysis (class D), based on the options listed in D5.

8.5 Valuation discipline

Only **proven and probable reserves** under KCMI/CRIRSCO should underpin financing. Measured, indicated and inferred resources should be excluded. Value should be built up in seven steps:



Figure 3: Why financeable value is far below in-situ value (illustration)



Class D illustration for a hypothetical deposit: 1,000 Mt reserve, US$80/t, 50 Mt/yr output, 15% state share, 20-year life, 8% discount rate, 1.5× coverage. Not an estimate for any real reserve.

In the illustration, a deposit with an in-situ value of US$80 bn supports only about **US$3.9 bn** of issuance, or about **4.9%** of in-situ value. The ratio changes with the assumptions, but the order of magnitude explains why a headline reserve value cannot be read as financing capacity.

Valuation and pledging rules

- No double pledging: the same revenue stream cannot back an NRF instrument and also be counted in the budget or pledged elsewhere.
- Independent competent persons and valuers, with annual revaluation and publication.

- Issuance capped at a fraction of present value, with a minimum coverage ratio (for example 1.5×) and a reserve account.
- Currency of the instrument matched to the currency of the revenue, or hedged.
- Revenue given up by future budgets disclosed alongside proceeds raised today.

8.6 Use of NRF proceeds

We propose five windows, each with its own rules, so that proceeds serve both saving and capability-building without blurring accountability.

Window	Purpose	Rules (proposed)
Stabilisation	Buffer the budget against commodity-price cycles	Withdrawal only by a published rule linked to price or revenue shortfall
Savings / future generations	Convert depleting resources into financial wealth	Long-term diversified portfolio; no domestic project lending
Industrial development	Equity and quasi-equity into anchor projects and downstreaming	Through Danantara/DDMF with financial and economic cases; additionality test
Supplier capability (DOCK-UP)	First-loss guarantees, certification co-funding, supply-chain finance support	Released only against verified gates; leverage private and bank credit
Social distribution	Benefit sharing with communities, including through village cooperatives	Through the budget or a separate programme window with its own audit; never from the investment windows

8.7 Governance

- **Legal basis** in law or regulation that defines the fund's ownership, mandate, deposit and withdrawal rules, and relationship to the budget.
- **Owner–manager separation**: the state, through a board, owns the fund and sets the mandate; professional managers manage assets under investment mandates (consistent with the Santiago Principles).
- **Independent audit** and annual publication of holdings, returns, flows to and from the budget, and instruments outstanding.
- **Risk limits** on concentration, currency, leverage and exposure to related parties.
- **Conflict rules** for officials who also hold roles in mining SOEs or regulators.

8.8 Recommended staged path

Stage	Timing	Content	Gate to next stage
0: Foundations	0–3 months after approval in principle	Legal rights review; reserves register under KCMI; valuation method; fiscal-risk treatment; governance design	Rights confirmed; valuation published; instrument classified as debt where applicable
1: Pilot	3–12 months	Royalty-linked or revenue-linked note on producing coal assets with proven reserves; capped size; domestic institutional investors	Successful issuance at a cost no higher than comparable SBN; covenants met
2: Broaden	12–24 months	Extend to strategic minerals; open the industrial-development and DOCK-UP windows	Independent evaluation of the pilot; register of pledged revenues
3: Mature fund	> 24 months	Stabilisation and savings windows; consider a Mineral Reserve Trust	Legislative basis complete; track record of returns and disclosure

09
Financing channel 2: sovereign project tokenisation

A project-level channel that turns verified state contracts into investable, milestone-released instruments. It is assessed through the PALCapital proposal for PT PAL.

9.1 The problem it addresses

PT PAL holds billions of dollars of state contracts but faces a working-capital gap [D6]:

- **Sovereign backlog**: two Scorpène Evolved submarines (US$2.16 bn), two Merah Putih AH-140 frigates (US$1.2 bn), plus LPDs and hospital ships, all MoD contracts with 60–96 month cycles.
- **Up-front cash outlay**: HY-80 steel, lithium-ion battery arrays, propulsion and combat systems must be paid before milestone payments arrive, creating a gap of **US$150–300 m per programme**.
- **Import dependence**: about 75% of materials are imported, so foreign-exchange and supply-chain risk is concentrated.
- **Slow conventional finance**: bank syndicates take 6–9 months, carry heavy covenants and cost 9–11% all-in.

9.2 The proposed instrument: PAL-SUB-1

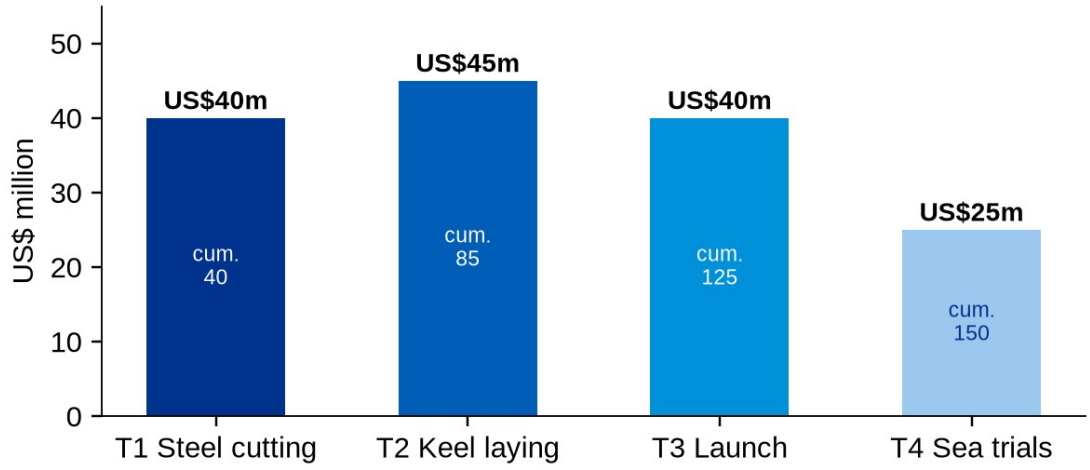
Feature	As proposed in D6
Legal nature	Yield-bearing debt note backed by a verified MoD purchase order; investor described as a secured creditor
Size and tenor	US$150 m (Phase 1, Scorpène Unit 1); 700 bps fixed yield; 720 days
Token standard and network	ERC-3643 (T-REX) permissioned security token on a private Hyperledger Besu network (IBFT 2.0); 4 validators (PAL ×2, Danantara, BKI), 3 needed to finalise
Privacy	Zero-knowledge proof of PO value and Ministry of Finance signature; classified specifications never on-chain
Custody and escrow	Fiat escrow at Bank Mandiri Custody under a tripartite deed PAL–Danantara–Mandiri; escrow bank determined by Danantara IC, MoF and OJK
Release control	Funds released in tranches direct to vendors after 2-of-2 sign-off by a BKI lead surveyor and a TNI AL officer, with geotagged photo, GPS and timestamp; IoT data supporting only in Phase 1; weld quality remains a manual BKI check
Repayment	All MoD payments (15% down payment and T1–T5) go to a Mandiri collection account; waterfall pays accrued yield first, remainder to PAL; at final delivery principal and final yield are paid and the token is burned
Investors	Danantara as anchor; Mandiri, BRI and sovereign funds as co-investors; 100 whitelisted wallets in Phase 1; no retail until Phase 3
Regulation and security	OJK IKD (digital innovation) sandbox; BSSN-compliant sovereign cloud; ISO 27001; FIPS 140-2 Level 3 HSM; read-only auditor nodes for MoD and OJK; upgrades via 3-day timelock and 3-of-5 multisig

Feature	As proposed in D6
Roadmap	Phase 1 Q4 2026: US$150 m pilot. Phase 2: US$300 m, frigate and submarine unit 2. Phase 3: US$1 bn+, full order book, retail IKD and secondary trading

9.3 Lifecycle and cash flows



Figure 4: PAL-SUB-1 milestone tranches



Source: D6. Tranches sum to US$150 m.

At 700 bps for 720 days, the yield on US$150 m is **US$20.7 m** (D6 rounds to US$21 m). D6's waterfall example, in which a T1 receipt of US$40 m pays US$5.17 m of accrued yield and forwards US$34.83 m to PAL, is consistent with about 180 days of accrual. D6 claims savings of US$8–12 m per vessel. The difference between a 10.5% bank cost and 7% on US$150 m over 720 days is about **US$10.4 m**, so the claim is arithmetically consistent. This checks arithmetic only. It is not a like-for-like cost comparison (Issue 4 below), and it excludes platform, custody, surveyor and legal costs.

9.4 Assessment

Strengths worth keeping

- **Milestone release direct to suppliers** is a strong control: funds cannot be diverted to operating accounts, and it fits DOCK-UP's principle that verified readiness unlocks money.
- **Independent physical verification** by BKI, with a defence officer co-signing, makes evidence of progress the trigger for payment.
- **Regulator visibility from day one** (read-only nodes for MoD and OJK) and phased, sandbox-gated scale-up.
- **Privacy by design** for classified programme data.
- **Collection-account waterfall** gives investors a clear priority and reduces the risk that payments are diverted.

Issues that must be resolved

Table 11: PALCapital issues and required resolution

#	Issue	Required resolution
1	**It is debt.** D6 describes the instrument as “not a grant, not debt” and also as a “yield-bearing debt note” whose investor is a “secured creditor”. In substance it is secured receivable financing.	Classify and report as a liability of PT PAL (or the issuing SPV) and include it in Danantara's group leverage and guarantee register [D1, D2].
2	**Assignment of state receivables.** Redirecting MoD payments to a collection account requires contractual rights and consent from MoD and the Ministry of Finance, and must fit budget-execution rules.	Written legal opinion; consent instruments from MoD/MoF before any token is minted.
3	**Completion and appropriation risk.** Investors are paid only if PAL reaches milestones and MoD pays on time. “Even if PAL defaults, future MoD payments still pay investors first” holds only while the programme continues.	Independent credit assessment covering programme delay, termination and budget timing; stress MoD payment delay of 180 days.
4	**Cost comparison not like-for-like.** 7% is compared with a 9–11% bank cost without stating currency, tenor or security, and without the platform's own costs (D6 adds 2–3% fees on the bank side only).	Compare same currency, tenor and security, including total cost of ownership of the platform, custody, surveyor and legal fees.
5	**Who is the investor?** With Danantara as anchor and state banks as co-investors, Phase 1 recycles state money. For comparison, the principal-weighted issue yield on DIM's June 2026 US$ notes was 5.65%, before fees and hedging [D1].	Treat Phase 1 as a control and technology pilot. Measure success by new domestic institutional investors in later phases, not by state-on-state funding.
6	**Is a blockchain needed?** Escrow, milestone release and audit trails can be built with conventional custody and digital registries. Tokenisation mainly adds value for distribution and secondary trading (Phase 3).	Make the pilot an explicit test: compare token-based and conventional escrow on cost, speed, control and risk.
7	**Concentration of roles.** Danantara is anchor investor, validator, member of the escrow-determining body and holder of an upgrade key.	Separate investment, validation and governance roles; give the independent validator and regulator decisive controls.
8	**Unverified claims.** Danantara “US$900 bn+ AUM”, “pre-cleared by BSSN and OJK”, the “IDDB 2024 sandbox precedent” and third-party market projections (US$2 tn by 2028; US$18.9 tn by 2033).	Treat as class C; obtain written confirmations before relying on them.
9	**Internal inconsistencies.** Go-live in 45 days versus 90 days; “no collateral” versus “MoD PO collateral”; “6–7% fixed” versus 700 bps; an unexplained “+200 bps”; a US$500 m value proof versus a US$150 m issue.	Issue a corrected term sheet before any approval.
10	**Import leakage.** With 75% of materials imported, most escrow releases will flow to foreign vendors.	Set a target share of escrow released to DOCK-UP-qualified domestic suppliers, rising by phase.

9.5 Conditions for approving the Phase 1 pilot

Six pre-conditions (recommended)

- **Legal**: opinion confirming the instrument's nature, plus MoD and MoF consent to assignment and payment redirection.
- **Fiscal and group risk**: recorded as a liability, included in group leverage limits and the guarantee register; independent credit assessment.
- **Economics**: like-for-like comparison, including platform total cost of ownership, against a conventional escrow-based facility.
- **Governance**: role separation for Danantara; independent validator and OJK sandbox registration in place; corrected term sheet.
- **Technology and security**: independent security audit, key-management and incident-recovery plan, BSSN sign-off.
- **Industrial linkage**: evidence standards for milestone release, BKI independence, and a target share of releases to DOCK-UP-qualified domestic suppliers.

9.6 From one pilot to a national instrument

If the pilot succeeds, the same structure can finance other state contracts. We propose a family of **Sovereign Project Receivable Notes** with common design rules (our proposal, class D):

Design rule	Content
Currency and pricing	Rupiah-denominated, priced off the SBN curve of matching tenor plus a disclosed spread, so that domestic pension funds, insurers and banks can hold them
Underlying	Contracted state payment obligations: defence contracts, availability payments, tipping fees or PPAs, or government programme receivables
Release	Milestone tranches released direct to suppliers after independent certification
Repayment	Collection account and published waterfall
Transparency	Regulator and auditor read access; public summary of releases by milestone and supplier category
Accounting	Recorded as debt of the issuer and included in group and fiscal-risk reporting

Substitution test for foreign loans (proposed). Before any new foreign loan (pinjaman luar negeri) is approved for a strategic programme, test whether domestic project notes could fund it at a comparable all-in cost after hedging.

Table 12: Possible applications by sector (proposed)

Sector	Possible underlying receivable	Main specific risk
Shipbuilding and defence	MoD contract milestone payments	Programme delay, classified data, imported content
Coastal and transport infrastructure	Availability payments under a PPP	Construction delay, land acquisition
Waste-to-energy	Tipping fees and PPA payments	Waste-volume and calorific risk; payment delays
Mineral downstreaming	Off-take receivables from creditworthy buyers	Commodity price; buyer concentration
Housing programme	Government programme payments to developers	Programme-design and appropriation risk

10
An integrated financing architecture

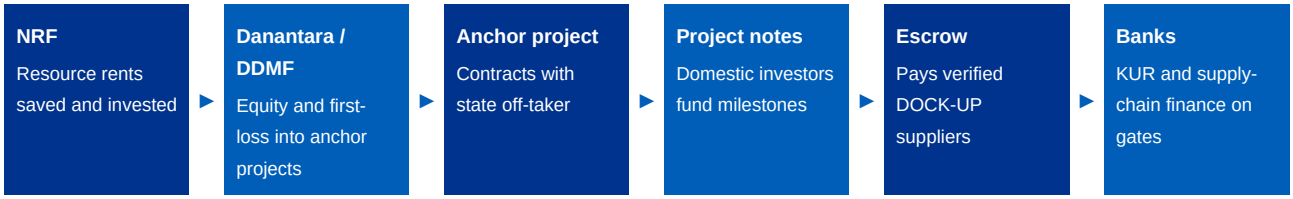
How the two channels fit together, how finance is matched to project and supplier gates, and how the whole is protected by one financing discipline.

10.1 Five financing principles

1. **Financing follows readiness.** No investment capital is committed before the project's capital decision (P3) or the supplier's relevant gate. Preparation funding before P3 is capped and treated as expense.
2. **Two cases for every project.** A financial case and an economic case; where public support is needed, it is quantified and approved explicitly [D1].
3. **Substance over label.** Any instrument with a repayment obligation is a liability, whether it is issued by an SPV, called a token, or kept off budget [D1, D2].
4. **No double pledging.** One register records every revenue stream pledged, every receivable assigned and every guarantee given.
5. **Stress the group, not the entity.** A project failure can cut SOE dividends, weaken a state bank and trigger a guarantee at the same time [D1].

10.2 How the two channels fit

	Natural Resource Fund	Sovereign project tokenisation
Level	Fund (upstream)	Project and supplier (downstream)
Source of value	State share of resource rents	Contracted state payment obligations
Typical use	Long-term equity, first-loss capital, stabilisation and savings	Construction and working capital, released by milestone
Horizon	Decades	1–5 years per instrument
Main risk	Commodity price, legal rights, fiscal substance	Completion, appropriation timing, technology and governance
Link to DOCK-UP	Funds the supplier window: guarantees and certification	Pays qualified suppliers directly from escrow



10.3 Sizing: what the two channels can and cannot fund

Table 13: Illustrative sources and uses

The brief asks how all projects will be funded. The stated portfolio is Rp2,091 trillion (class C, unreconciled). The table compares it with illustrative capacity from each identifiable source over five years. All capacity

figures are **class D illustrations** whose assumptions are stated, so they can be replaced with approved numbers.

Source	Illustrative five-year capacity	Basis and assumptions
Danantara strategic allocation	≈ Rp578 tn	2026 plan of up to US$14 bn with ~50% to strategic projects [D3 p.1], held at US$7 bn a year for five years; Rp16,500/US$
NRF Stage 1 revenue-linked notes	≈ Rp118 tn (≈ Rp83 tn if coal prices fall 30%)	20% of 2024 coal royalty (~Rp126.4 tn [D5]) = Rp25.3 tn a year ring-fenced for 10 years; PV at 7% = Rp178 tn; 1.5× coverage. This revenue is given up by future budgets
Sovereign project receivable notes (identified)	≈ Rp24 tn	PALCapital roadmap US$150 m + US$300 m + US$1 bn [D6]; other sectors not yet identified
Subtotal: the two new channels (NRF + project notes)	**≈ Rp142 tn (6.8%)**	Danantara's allocation adds ≈ 27.6%
Remaining to be funded	**≈ Rp1,371 tn (65.6%)**	Project-level debt (banks, bonds, export credit), SOE balance sheets, private and foreign co-investors, and the state budget

Class D illustration. Percentages are of Rp2,091 tn. The strategic half of Danantara's plan is not reserved for the 40 projects, and the portfolio's spending period is not known, so the five-year comparison is indicative only. The NRF line assumes the Government proceeds and Stage 0 confirms legal rights.

The result changes the financing question. The two new channels are useful complements: the NRF for patient equity and first-loss capital, and project notes for milestone-released working capital. But they cannot fund the portfolio. The decisive lever is **mobilisation**. If the portfolio is financed with 30% equity and 70% debt, the equity required is about Rp627 trillion. That is broadly the order of Danantara's illustrative strategic allocation plus an NRF industrial window. About Rp1,464 trillion would then be needed as debt, from banks, bond markets, export-credit agencies and project notes. Three conclusions follow:

1. Danantara's capital should be used mainly as equity and first-loss capital that unlocks debt and co-investment, not to fund projects outright.
2. Mobilisation should be measured with the attributable-mobilisation metric (I10).
3. The register reconciliation (Section 4.6) must establish the timing of capital needs by year, because a five-year comparison hides peaks.

Timing matters too. The NRF's industrial and DOCK-UP windows open only at Stage 2, 12–24 months after approval in principle and only if earlier stages succeed (Section 8.8). Near-term needs must therefore be met from Danantara's balance sheet and from bank finance.

10.4 Finance matched to gates

Projects and suppliers move through their gates on separate clocks. Serial supply (G3), for example, is usually needed during construction (P4), not at commissioning. The two ladders are therefore shown separately.

Project gate	Project instrument
Preparation (P0–P2)	DDMF project-preparation funds and grants for studies (capped, expensed)
Capital decision (P3)	Danantara equity; NRF industrial window; viability-gap support where economically justified
Construction (P4)	Sovereign project receivable notes; bank debt; export credit

Project gate	Project instrument
Commissioning (P5)	Refinancing into long-term bonds
Operation (P6)	Asset recycling to private investors

Supplier gate	Supplier instrument (DOCK-UP)
Entry	Companion engineers and training; research grants for explorers
G1	Matching grant; shared facilities
G2	Supply-chain finance; KUR
G3	Investment credit with first-loss guarantee (guarantee provider; NRF supplier window once open)
G4	Corporate venture capital; R&D super-deduction (PP 45/2019); LPEI export finance

10.5 Financial return versus economic return

Table 14: Financial versus economic case (D1 worked example)

D1's worked example shows why both cases are needed and why public support must be explicit. A hypothetical project costs Rp100 bn and yields Rp15 bn a year for ten years, plus Rp6 bn a year of external economic benefit.

Measure	Result	Interpretation
Financial NPV at 12%	−Rp15.25 bn	Below the commercial hurdle
Project financial IRR	8.14%	Not an equity IRR; financing cash flows excluded
Economic NPV at 9%	Rp34.77 bn	Positive only if the external benefit is defensible
Economic IRR	16.40%	Includes benefits unavailable to the investor
Viability-gap grant	Rp15.25 bn	Makes investor NPV zero; a public cost and a transfer

A positive economic result cannot be presented as investor profit, and a grant improves the investor's case without creating economic benefit. ADB's 9% economic-return reference (6% for specified social projects) is a methodological precedent, not adopted Indonesian policy [D1/S26].

10.6 Stress tests

All shocks below are illustrative starting assumptions, to be calibrated to each instrument and approved risk appetite. The first seven are from D1; the last four are our additions for the two new channels.

Risk	Proposed test	Decision response
Currency / refinancing	IDR −15%; refinancing spread +200 bps on debt actually repricing	Resize or hedge; preserve liquidity; revise distribution policy
Industrial cycle	Product price −20%; feed/power +15%; utilisation below plan	Re-test margin, covenants, offtake; stage capex
Construction	Capex +20%; 12-month delay; contractor failure; land delay	Fund contingencies; independent engineer; redesign or stop

Risk	Proposed test	Decision response
Fiscal / public service	Compensation paid 180 days late; guarantees called; subsidy cut	Disclose contingent support; funded service contracts; liquidity buffers
Bank credit	Credit migration; higher cost of risk; correlated SOE exposure	Assess CET1 and liquidity after losses; cap concentrated lending
Technology / market	Pilot yield not replicated; qualification delayed; IP inadequate	Link next funding to accepted capability milestones
Environmental / social	Baseline emissions overstated; permit or safety failure	Re-estimate benefits; suspend where binding conditions fail
NRF commodity	Coal price −30% for three years; output −20%	Coverage ratio test; draw on reserve account; halt new issuance
NRF legal	Challenge to rights over pledged revenue	Pre-issuance legal opinion; issuance only on undisputed assets
Project-note payment	MoD/state payment 180 days late; one milestone slips 9 months	Liquidity line; investor communication; no early release
Platform incident	Key compromise or smart-contract fault	Freeze via timelocked governance; fall back to the fiat escrow ledger

Disclosure

Publish a register of all resource-backed and receivable-backed instruments: issuer, amount, currency, tenor, recourse, revenue pledged, present value of revenue given up, and coverage ratio. Do not classify a structure as free or non-debt because it is outside a budget line or issued by an SPV [D1].

11
Review of the general-contractor SOEs

The state-owned construction companies (BUMN Karya) moved from contracting into infrastructure investment financed with debt. This chapter sets out their condition and a path back to a disciplined, healthy contractor model.

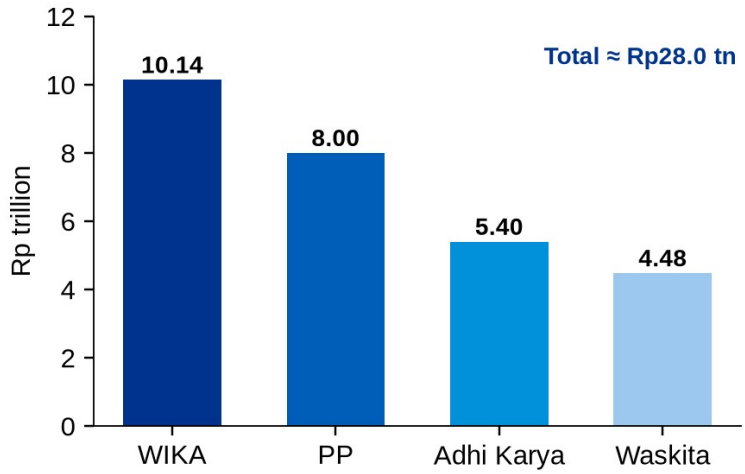
11.1 Condition

≈ Rp28 tn
Estimated combined 2025 losses of WIKA, PP, Adhi Karya and Waskita

≈ −45%
Infrastructure budget cited in D3: Rp110.95 tn in the prior year → Rp61.38 tn in 2025 (−44.7%)

Rp10.14 tn
Largest single estimated loss (WIKA)

Figure 5: Estimated 2025 losses of the four construction SOEs



Source: D3 p.3. Evidence class C (estimate); confirm against audited FY2025 statements. The four values sum to Rp28.02 tn.

Audited figures must replace the estimates

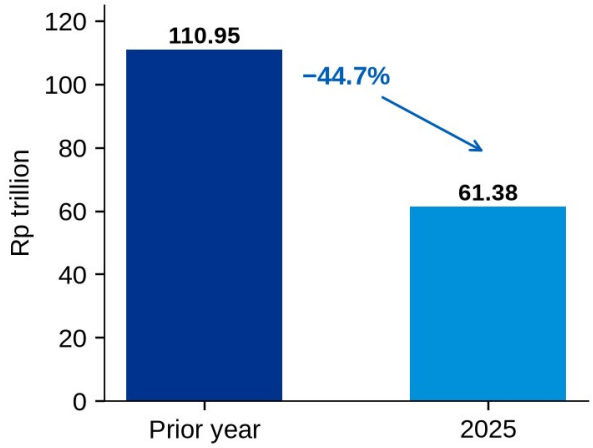
The loss figures are estimates (class C). The four companies are publicly listed, and D1 reports that SOEs completed their FY2025 statements by July 2026 [D1/S02]. The diagnostic should therefore start from audited FY2025 accounts and fill the template below for each company, including Hutama Karya and the smaller construction SOEs if they are in scope. Any consolidation plan for the construction SOEs should be tested against the same data.

Table 15: Diagnostic data template per construction SOE (to be populated)

Area	Data required
Results	Revenue, EBITDA and net profit or loss, FY2023–FY2025, audited; restatements
Debt and liquidity	Total and interest-bearing debt; net debt/equity; interest coverage; cash runway; maturity wall; covenants

Area	Data required
Receivables	Days sales outstanding; amounts owed by government, by SOE owners and by others; ageing; disputed amounts
Order book	New contracts and backlog by owner type and sector; margin at bid; share of loss-making contracts
Legacy assets	Concessions, property and other non-core assets at book and fair value; encumbrances; funding needs
History of support	Restructurings, PKPU, capital injections (PMN) and their conditions

Figure 6: Infrastructure budget before and in 2025



Source: D3 p.3. The deck rounds the fall to 45%.

Main problems

- Many projects with low margins, or assigned for policy objectives without full commercial compensation or funding.
- Toll roads and other infrastructure financed with corporate debt.
- Delayed receivables and slow payment from government and project owners.
- Cost overruns and project delays.
- High leverage, low asset quality and negative cash flow.
- Governance that did not sufficiently protect commercial discipline.

Source: D3 pp.3 and 5. D3 does not name the budget line or the prior year; the figures should be confirmed against the state budget documents.

Root causes

Cause	Mechanism
Aggressive infrastructure expansion	Many toll-road, rail, airport, dam and port projects taken on in a short period
Low construction margins	Intense competition and policy-assigned projects without full compensation
Late payment by government and SOE owners	Receivables became a funding problem
Turnkey and EPC risk	Contractor carries design, procurement, construction and commissioning risk
Governance and incentives	Success measured by project completion, not profitability

The consequences are negative cash flow, high leverage, poor asset quality, declining capacity to take on projects, and risk to the companies' continuity [D3 p.3].

11.2 Current model versus the model required

Current model: infrastructure investment conglomerate	Target model: disciplined construction company
Main contractor and EPC contractor	Focus on the core construction business
Project developer	Selective: only projects with healthy margins
Toll-road investor and property developer	No excessive ownership of long-term assets
Owner of infrastructure assets	Project funding from the owner, not from company debt
Supplier through subsidiaries	Strong project risk management
Financier of government projects	Professional and independent governance
Instrument for accelerating national infrastructure	High operational capability and productivity; sustainable growth and profitability

Source: D3 p.3.

Assets that became burdens

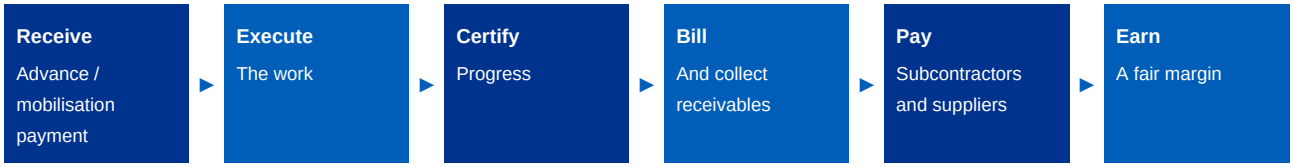
Toll roads, residential property, airports, power plants, ports, mining concessions, hotels, industrial estates and factories. D3 recommends that these assets be placed in a **separate investment company with separate funding and governance** [D3 p.3].

11.3 Direction of transformation



The healthy contractor's cash cycle

A healthy construction company is paid before it spends and bills as it builds. The six principles from D3 form a cycle that every bid and every project should be tested against:



11.4 Recommendations

#	Recommendation	Content
1	**Diagnostic per company (Day 30–90)**	Audited FY2025 results; backlog quality and margin; receivables ageing by owner; debt maturities and covenants; legacy-asset register with valuations; claims and variation orders outstanding; cost-to-complete on loss-making projects
2	**Separate legacy assets after the diagnostic**	Transfer concessions and non-core assets to an asset vehicle with its own funding and governance; value at transfer with legacy liabilities separated [D1]; do not book transfers as returns

#	Recommendation	Content
3	**Bid discipline**	Bid/no-bid gate with minimum risk-adjusted margin, payment security (owner funding, advance payment, payment guarantees) and working-capital test; no corporate-debt financing of owners' projects
4	**Cash protection**	Escalation of overdue receivables from government and SOE owners; funded public-service contracts where SOEs act on assignment [D1]; collection-account and milestone mechanisms where suitable (Chapter 9)
5	**Project control backbone**	Common WBS and cost codes; commitment tracking; budget, quantity and timing checks at purchase request; three-way match; earned-value reporting; deployed through the AI operating system (Chapter 12)
6	**Role in anchor projects**	Act as disciplined EPC contractors and as integrators for DOCK-UP packages in housing and coastal infrastructure, with owner funding
7	**Structure decided on evidence**	Decide on consolidation or specialisation only after the diagnostic; fewer legal entities is not a benefit in itself [D1]

How to measure recovery

Metric	Why it matters
Incremental economic profit (O02)	Tests value creation above the cost of capital against a without-intervention path
Operating cash conversion (O04) and days sales outstanding	Shows whether margin turns into cash; does not reward overdue suppliers
Net debt / EBITDA (O08); interest coverage; net debt / equity; liquidity runway	Solvency and refinancing headroom. Where EBITDA is negative, ratios based on it are meaningless, so use interest coverage, net debt to equity and months of liquidity instead
Backlog margin at bid versus at completion	Tests bid discipline and estimating quality
Cost and schedule performance indices (CPI, SPI)	Early warning of overruns at project level
Claims and variations recovered / submitted	Commercial management capability
Verified net transformation benefits (O09)	Realised cash benefits net of implementation costs

12

Improving business management with multi-agentic AI

An AI-based project operating system that protects cash, detects losses earlier, raises productivity and coordinates the SOE construction portfolio. It is built on existing systems, with humans accountable for every material decision.

12.1 What is proposed

D3 proposes **not a chatbot but an AI-based operating system for projects** that oversees work from individual projects up to the portfolio. Specialist AI agents analyse data from existing systems, predict risks and recommend actions; orchestrator agents coordinate them, resolve conflicting outputs and escalate high-risk decisions to people [D3 pp.4–6].

Pillar 1
Integrated data
One source of truth across ERP, BIM, procurement, contracts, finance and field data.

Pillar 2
AI agents
Analysis and smart recommendations by specialist agents.

Pillar 3
Collaboration
Across SOEs and stakeholders, from project manager to Danantara.

Pillar 4
Tangible impact
Productivity, industrial resilience and national value.

12.2 Target architecture

Table 16: Layered architecture of the AI project operating system

Layer	Role	Components
Danantara Portfolio Control Tower	Real-time, data-driven monitoring and strategic direction of the construction SOE portfolio	Portfolio orchestrator; capital allocation by risk and strategic value; SOE benchmarking (performance, margin, productivity); turnaround projects; strategic project priorities aligned with the national agenda; domestic industry impact (TKDN, capability development)
SOE Enterprise Orchestrator	Coordination across projects for resources, finance and execution	CEO, CFO and COO cockpits and Risk & Compliance; functions for strategy and business development, finance and treasury, operations and projects, risk and compliance, HR and capabilities, consolidation and reporting
Corporate AI agent groups	Specialist agents by business function	**Commercial**: bid/no-bid, tender estimation, contract risk, claims, client payments, business development. **Finance**: cash and liquidity, receivables and collection, payables, treasury and funding, project profitability, consolidation. **Delivery**: project control, cost, schedule, procurement, supplier management, equipment and workforce, quality, HSE
Project Control Tower (per project)	Agent coordination at project level for cost, schedule and quality	Project manager, cost, schedule, claims, procurement, supplier-risk, equipment, workforce, quality, safety, sustainability and BIM agents

Layer	Role	Components
Field intelligence	Real-time, evidence-based data from site	Mobile site app, drones, CCTV, IoT sensors, digital survey, BIM/digital twin, SCADA
Enterprise data platform	Central data layer with consistent master data for all agents, integrated with existing systems	ERP (finance, project, procurement), e-procurement, contract management, BIM/CDE, scheduling (Primavera P6), HR, finance, legal, safety and quality, equipment, IoT, external data (market prices)

Source: D3 pp.5–6.

Master data: the foundation

All agents share one set of identifiers: **Project ID, WBS, cost code, activity code, material code, supplier ID, contract ID, variation-order ID, payment-certificate ID and asset ID** [D3 p.6]. Without consistent master data, agents produce confident answers from inconsistent numbers. Master-data work is therefore month-one work, not an afterthought.

Data and interaction flow



12.3 Priority agents

Agent	Functions
Portfolio risk agent	Ranks projects and subsidiaries by risk (cost, schedule, payments, receivables); early detection of potential losses; Green / Amber / Red / Black classification; recommends recovery plans
Bid / no-bid agent	Evaluates tender feasibility; analyses historical margins, contract risk and working-capital needs; simulates base, downside and severe-downside scenarios; recommends go / no-go
Cost estimating agent	Independent estimate from historical data and market prices; accounts for productivity, logistics, ground and weather conditions; compares with contract and changes; detects cost-risk
Receivables and payment agent	Monitors every invoice and payment certificate; detects delays and missing documents; reconciles with certification; predicts payment default; recommends collection acceleration and cash preservation
Schedule and progress agent	Monitors physical progress against plan; analyses delays and causes; assesses impact on cost and margin; integrates field data (photos, drones, IoT); recommends acceleration
Procurement and supplier agent	Analyses suppliers and market prices; detects supplier risk; optimises procurement strategy; monitors contracts and deliveries; recommends substitution and cost efficiency

Source: D3 p.5.

12.4 Embedding project-control discipline

The agents are only as good as the control logic they enforce. We recommend that the first release encodes the standard EPC budget chain (estimate → control budget → commitment → actual → payment, with progress billing and collection on the revenue side) as explicit rules that agents check continuously:

Control	What the agent checks	Agent
One cost-code backbone	Every budget line, PR, PO, goods receipt and invoice carries the same WBS × PBS × cost-element code	Cost
Commitment tracking	Budget versus committed versus actual versus estimate at completion per control account, every week	Cost / Portfolio risk
Three checks at purchase request	Budget (money left on the WBS account), quantity (within the PBS item's material take-off balance) and timing (arrives before the required-on-site date)	Procurement
Three-way match	PO = goods receipt or certificate = invoice before payment; flags POs raised after work started	Receivables & payment / Procurement
Earned value	CPI and SPI per control account; EAC from bottom-up forecast compared with AC + (BAC − EV)/CPI	Schedule & progress / Cost
Revenue protection	Unbilled work, unapproved variation orders, retention and advance-payment recovery; ageing by owner	Receivables & payment

12.5 Human in the loop

Agents analyse and recommend; people decide. Approval is required for material decisions, people intervene on complex cases, and accountability is preserved [D3 p.6]. We propose four autonomy levels with explicit limits:

Level	Agent may	Examples	Never permitted
L0 Inform	Report and alert	Overdue receivable alert; CPI below threshold	–
L1 Recommend	Propose an action with evidence and alternatives	Bid/no-bid recommendation; recovery plan	–
L2 Prepare	Draft a transaction or document for human approval	Draft collection letter; draft change-order assessment	Submitting without approval
L3 Execute within limits	Perform low-risk, reversible actions inside set limits	Send reminders; update dashboards; open tickets	Approving payments, bids, contracts, budgets or personnel decisions

12.6 Three-month MVP

	Month 1: foundation and data integration	Month 2: six priority agents	Month 3: integration, pilot and go-live
Work	Finalise scope and use cases; read-only integration with existing ERP; enterprise data layer and master data; infrastructure, security and access; multi-agent framework and tool gateway; recruit and onboard team; initial dashboard design	Develop and test six agents. D3 lists: (1) project document, (2) cost, (3) cash and receivables, (4) procurement, (5) schedule, (6) project-manager orchestrator. We recommend replacing the project-document agent with the **bid/no-bid agent**, because Chapter 11 traces losses to bidding. Integrate with ERP data; test and validate with key users; iterate	Full integration across agents; executive dashboards (Danantara, CEO, CFO, project manager); pilot on 2–3 priority projects; user acceptance testing; training; documentation, SOPs and handover; evaluate and plan next phase
Milestone	Team onboarded; ERP data layer ready; agent architecture selected; initial dashboard; sample data from 1–2 projects	Six agents built; internal integration tested; first insights from real project data; iteration on user feedback	Pilot on 2–3 projects running; executive dashboard ready; UAT and training complete; SOPs documented; next-phase plan evaluated

Source: D3 p.7, with our recommended change to the agent list. D3's six MVP agents differ from its six target agents in Section 12.3; the portfolio-risk agent should follow in Phase 2. D3's Gantt chart labels twelve columns M1–M12 within three months; we read these as weeks.

Table 17: MVP team

Technical team (14)	No.	Business and domain team (6–8)	No.
Project lead	1	Project control expert (WBS, cost, schedule, risk)	1–2
AI / agent engineers	3	Finance / CFO expert (cash, receivables, payables, funding)	1
Backend engineers	3	Procurement expert	1
Data engineers	2	Contract / claims expert	1
AI/ML engineer	1	ERP functional specialist	1
Frontend engineers	2	Business analyst / product owner	1
DevOps / security	1	Data governance specialist	1
QA engineer	1		

Source: D3 pp.7–8. D3 p.7 counts the project lead within the 14; D3 p.8 lists 13 specialist roles plus a separate security/IAM engineer and no project lead. The split should be confirmed; total remains 14.

12.7 Computing resources and scaling

Table 18: Scaling path

Phase	Timing (from MVP start)	Scope	GPU
1 Pilot	0–3 months	1 SOE, 2–3 projects, 6 main agents, limited users, validate architecture and data model	4 GPU (1 site)
2	6–12 months	2–3 SOEs, 20–50 projects, more agents and users, high availability	8 GPU (2 sites / HA)

Phase	Timing (from MVP start)	Scope	GPU
3	12–24 months	All construction SOEs, 100+ projects, Danantara portfolio agents, large-scale computer vision, selected internal and external users	16 GPU (cluster)
4 National	> 24 months	All construction SOEs, hundreds of projects, full SOE analytics, industrial ecosystem integration, disaster recovery, services for national stakeholders	16–24+ GPU (multi-site cluster)

Table 19: Phase 1 infrastructure

Component (Phase 1)	Qty	Specification	Function
GPU server (LLM inference)	1–2	4 × NVIDIA L40S 48 GB per server; 32–64 CPU cores; 256–512 GB RAM; 4–8 TB NVMe; 10/25 GbE	LLM serving (30–40B and 7–14B models), agent inference, embeddings, monitoring
Data platform server	1	32–64 cores; 256 GB RAM; 8–16 TB NVMe	Database, lakehouse, vector DB, ETL, master data, knowledge graph, API gateway, workflow, audit log
Document processing server	1	16–32 cores; 64–128 GB RAM; optional 1 GPU (L4/L40S)	OCR of contracts, drawings and reports; computer vision for progress; document classification
Storage / backup	1	20–50 TB NAS / object storage	Documents, BIM, images, video; backup and disaster recovery
Network	–	10/25 GbE	Server interconnect and internal access

Table 20: Model classes

Model type	Examples named in D3	GPU estimate	Main use
Main LLM (30–40B)	Llama 3.1 70B (4-bit), Qwen 2.5 32B, Mistral Large	1–2 GPU (48 GB)	Agent reasoning, document analysis, executive summaries, recommendations
Small LLM (7–14B)	Llama 3.1 8B, Qwen 2.5 7B	1 GPU (48 GB)	Classification, routing, information extraction, light tasks between agents
Embedding model	bge-large, E5-large	1 GPU (48 GB)	Vector search, document RAG
Computer vision (optional)	YOLO, SAM, VideoMAE	1 GPU (48 GB)	Progress analysis; object and activity detection

Source: D3 p.8. D3 groups Llama 3.1 70B (4-bit) under the 30–40B class; it is a 70B model quantised to fit and should be load-tested. Model choices date quickly: select current open-weight models with suitable commercial licences at procurement.

The end state is a **hub-and-spoke** design: a Danantara AI platform (LLM gateway, agent platform, common data model, identity and access management, model management, portfolio control tower) as the hub, with AI control towers at WIKA, PP, Waskita, Adhi Karya and other construction SOEs as spokes. The platform team of about nine (3 AI/agent, 2 data, 2 backend, 1 DevOps/cloud, 1 security/IAM) builds the common platform and portfolio agents; each SOE pilot team of about five (2 frontend, 1 AI/ML, 1 QA, 1 integration engineer) integrates its ERP and project systems and runs user acceptance and training [D3 p.8].

Beyond construction

Once proven, the same platform and master data can host the DOCK-UP readiness register and supplier portal, the data for the three scorecards, the NRF reserves and pledged-revenue register, and the evidence

trail for milestone-released project notes. One platform then supports the capability, financing and performance chain described in this study.

12.8 Governance of the AI system

Area	Requirement (proposed)
Data sovereignty	Local infrastructure and open-weight models, as D3 proposes; classified or sensitive data never leaves controlled environments
Security	Role-based access, multi-tenant isolation between SOEs, audit logs of every agent action, BSSN-aligned controls
Model risk	Evaluation sets per agent; guardrails; every recommendation cites the source records it used; periodic accuracy review against outcomes
Read-only first	Phase 1 integration with ERP is read-only; write-back only after controls are proven
Accountability	Each agent has a named business owner; human approval for all material decisions (Section 12.5)
Benefits	Measured as verified net benefits after implementation and running costs (O09), not as hours saved or reports generated
Vendor independence	Open interfaces and portable models to avoid lock-in

Expected results and how to measure them

Expected result (D3)	Proposed measurable indicator
More contracts with healthy margins	Share of new contract value bid at or above minimum risk-adjusted margin; margin at completion versus at bid
Faster collection of receivables	Days sales outstanding; overdue receivables by owner; unbilled work
Fewer problem projects and losses	Share of projects Red/Black; time from first warning to recovery plan; loss provisions
Better cash flow and balance-sheet health	Operating cash conversion; net debt/EBITDA
Adoption (proposed)	Weekly active users by role; share of decisions taken with agent evidence attached

Business case, feasibility and change management

D3 does not cost the programme. A business case should be prepared before Phase 2. Its benefit side can be sized simply. For an illustrative construction SOE with Rp20 trillion of annual revenue, cutting days sales outstanding by 10 days releases about Rp0.55 trillion of cash, which saves about Rp55 billion a year at a 10% cost of debt. Avoiding one loss-making Rp1 trillion contract bid at −10% margin avoids a loss of about Rp100 billion. These are class D illustrations. On these orders of magnitude, an MVP of the size described is likely to pay back quickly if even one such effect is realised and verified, to be confirmed when the MVP is costed.

- **Feasibility.** Recruiting 14 specialists and procuring GPU servers through SOE procurement in month 1 is ambitious. Use existing sovereign-cloud or SOE infrastructure for the MVP where possible, and procure dedicated hardware for Phase 2.
- **Change management.** Name a business owner per agent. Require decision logs that record whether agent evidence was used. Align incentives with the cash and margin indicators above. Consult HR and worker representatives where roles change.

13
Governance, organisation and decision rights

Who integrates the portfolio, who decides, and how independence of risk, legal and audit is protected while projects are accelerated.

13.1 Four organisational options

Option	Strengths	Limitations	Assessment
A Strengthen CTO and DDMF	Fast; uses existing functions; suits development and technology projects	Mandate across DAM and DIM and corporate results needs a clear charter	Viable if almost the whole portfolio sits within DDMF's mandate
B Strategic Industry unit at BPI	Integrates the 40 projects across arms; one owner of portfolio management	Risk of becoming another meeting layer without data access and blocker-resolution authority	**Recommended initial option**
C Formal Deputy Head leads the arm	Senior executive standing can strengthen coordination and continuity	Needs legal certainty on the position, hierarchy and authority; risk of centralisation or conflict with COO/CIO/CTO	Development option once the need for authority is proven
D New industrial holding or body outside BPI	May suit ownership of specific platforms or a distinct cross-sector mandate	Complex asset transfers, funding, shareholding and regulation; may fragment the portfolio	Only if the need for a separate legal entity is proven

Source: D2 §6.

Option B comes first because the gap is **portfolio integration and blocker resolution**; capital and project-development functions already exist. A BPI-level unit can manage cross-arm dependencies without waiting for projects to be moved or new companies to be formed, and projects already under way keep their lawful contract and decision paths [D2].

13.2 NITO: implementing the National Industrial Transformation Office as a unit

D3 proposes a **Danantara National Industrial Transformation Office (NITO)** with eight functions. We recommend implementing NITO as the Option B unit inside BPI, not as a holding or a new legal entity, and mapping its functions onto the structure D2 proposes:

NITO function (D3)	Placement in the unit (D2 structure)	Note
Industrial strategy and roadmap	MD Strategic Industry Development	Integrates Industrialization, Chief Economist, Stakeholders and Global Relations
Project development (pre-FEED, FEED, FID)	MD Engineering and Execution	Uses Technology Advancement and Strategic Technology Initiatives; each project keeps its own engineering lead
Technology and IP	MD Engineering and Execution with the CTO	CTO provides independent technical review of vendor claims

NITO function (D3)	Placement in the unit (D2 structure)	Note
Supply-chain development (DOCK-UP)	MD Strategic Industry Development; cluster leaders	Runs the DOCK-UP office and readiness register
SOE demand integration	MD Portfolio and Project Delivery; cluster leaders	Aggregates anchor-project demand into localisation plans
Industrial financing	Prepared by the unit; decided by CIO/DIM and DDMF	The unit prepares packages; it does not approve capital
Performance office	MD Portfolio and Project Delivery	Maintains register, scorecards and benefits tracking
Independent assurance (D3: "asuransi independen")	**Outside the unit**: Internal Audit and independent evaluation	A unit cannot assure its own work; this function must report independently

13.3 Unit structure

Role	Main responsibility	Relationship with existing functions
Head of Strategic Industry and Project Delivery	Integrated priorities, dependency resolution, quality of decision packages, team capacity, results reporting	Reports to the Head of the Executive Body; no universal approval authority
MD Portfolio and Project Delivery	Register of 40 projects, integrated schedule, blockers, data quality, decision reports and benefits	Office of the Board, Digital Solutions and Finance support systems and the decision trail
MD Strategic Industry Development	Value-chain architecture, markets, offtake, partners, import substitution, supplier capacity, national benefits	Integrates Industrialization, Chief Economist, Stakeholders and Global Relations
MD Engineering and Execution	Study readiness, engineering, contract strategy, constructability, cost, commissioning and technical evaluation	Uses Technology Advancement and Strategic Technology Initiatives
Four cluster leaders	Resolve dependencies within a cluster; map raw materials, markets, technology and shared facilities	Matrix teams from SOEs, holdings and BPI functions by assignment

Staffing starts with secondment from existing functions. The number of MDs, project directors and experts should follow the number of active projects and their stages; no one leader should manage 40 projects day to day. Staff plans should fill proven gaps; this study does not assume headcount or overhead before an inventory [D2].

13.4 Working relationships

Function	Retains responsibility for	Relationship with the unit
COO and DAM	Management and transformation of portfolio operations	Assigns sponsors and SOE operating capability; agrees asset boundaries and operational readiness
CIO and DIM	Commercial investment testing, capital allocation, investment partnerships	Reviews the financial case and funding structure within the same decision package
CTO	Technology architecture, scale readiness, IP, industrial capability strategy	Independent technical review of vendor claims; technology selection input

Function	Retains responsibility for	Relationship with the unit
DDMF	Preparation, structuring, capital mobilisation and delivery of development projects	The unit manages dependencies across the portfolio; DDMF delivers projects assigned to it
Finance	Liquidity, cost of capital, accounting, exposure, consolidation	Tests capital sources and financing; supports project cost monitoring
Risk and Legal	Risk opinions, compliance and authority, with freedom to object	Do not report substantively to project promoters; opinions recorded in decision minutes
Internal Audit	Assurance over controls and follow-up of findings	Accesses information and reports through the lawful audit line; never an investment approver

The COO, CIO and CTO also head DAM, DIM and DDMF respectively. These dual capacities should be handled by separating decision documents and managing conflicts when interests diverge. One information package should serve several reviews, and each object should receive one decision from the entity with authority; coordination forums must not become a universal approval layer [D2].

13.5 Decision rights for key decisions

Decision object	Who decides	Role of the unit
Portfolio priorities	Unit head drafts; Executive Body adoption by the Head within authority	Integrates proposals and shows resource conflicts
Executive Body plan and changes	Head proposes; Supervisory Board and President approve as regulations require	Prepares projects and budgets for the plan; does not replace approvals
Investment by a company	Board of directors, GMS or other organ per articles, limits of authority and regulation	Tests readiness and dependencies; not approval of all group investment
Borrowing or collateral by the Executive Body	Presidential approval and Supervisory Board authority under PP; executed by the Body when complete	Prepares needs and alternatives; Finance and Legal set the tested structure
Guarantees by the Executive Body to investment holdings	Supervisory Board approval and related Presidential approval checked under PP	Never issues guarantee commitments on the Body's behalf
Construction contracts and changes	Authorised organ or officer of the implementing company	MD Engineering reviews; project director controls execution
IP rights and standards	Agreements and company organs decide IP rights; state authorities set mandatory standards (including TKDN)	Negotiates IP strategy and technical requirements; does not set national standards

Source: D2 §12.

13.6 RASCI for the 24 core activities

R = responsible; A = single owner of the final result; S = support; C = consulted; I = informed. The RASCI is a proposal for work allocation and creates no new legal authority. Each of the 24 activities has exactly one A [D2 §13–15].

Code	Role	Code	Role
KP	Head of the Executive Body	CI	CIO or assigned DIM function
SI	Head of Strategic Industry at BPI	CT	CTO or assigned technology function

Code	Role	Code	Role
PD	MD Portfolio and Project Delivery	DD	DDMF or assigned MD Development Investment
ID	MD Strategic Industry Development	RS	Independent risk function
EN	MD Engineering and Execution	LC	Legal and Compliance
SP	Corporate sponsor or assigned project company	FI	Finance
PJ	Project director at the implementing company	IA	Internal Audit
CO	COO or assigned DAM function	DW / PR	Supervisory Board / President
		OG	Lawful company organ for the decision object

Activity	A	R	S	C	I
S01 Reconcile the 40-project register	PD	PD SP	FI	LC CO CI DD	SI KP
S02 Prepare industry map and priorities	SI	ID PD	EN	CO CI CT DD RS FI	KP
S03 Set Executive Body priorities per mandate	KP	SI	PD	CO CI CT DD RS LC FI	DW
S04 Project feasibility and readiness package	SP	PJ ID EN	PD	CI DD CT RS LC FI	SI
S05 Technology and IP opinion	CT	CT EN	ID	SP LC	SI CI DD
S06 Independent risk opinion	RS	RS	FI	LC CT SP	KP SI
S07 Legal opinion and authorisation path	LC	LC	PD	RS FI SP	KP SI
S08 Draft financing structure	CI	CI FI DD	SP	RS LC SI	KP
S09 Portfolio recommendation package	SI	PD ID EN	SP	CO CI CT DD RS LC FI	KP
S10 Executive Body decision within authority	KP	KP	PD	SI RS LC FI	DW SP
S11 Presidential approval where required	PR	DW	KP PD	LC FI RS	SI
S12 Supervisory Board approval where required	DW	DW	KP PD	LC FI RS	PR SI SP
S13 Company investment decision per articles	OG	SP	CI DD FI	RS LC CT SI	KP PD
S14 Contract effectiveness and financial close	OG	SP PJ	FI CI DD	RS LC EN	SI PD
S15 Construction as assigned	PJ	PJ EN	PD	SP RS LC FI	SI
S16 Material contract or cost change	OG	SP PJ	EN FI	RS LC CI DD SI	KP PD
S17 Resolve internal portfolio issues	SI	PD	ID EN SP	CO CI CT DD LC	KP
S18 Escalate cross-agency issues to lawful forum	KP	SI	PD	LC SP CO CT	DW
S19 Company commissioning acceptance	OG	SP PJ	EN CT	RS LC	SI PD
S20 Operation and asset performance	SP	SP	CO PJ	FI RS CT	SI PD
S21 Consolidate portfolio benefits and results	PD	PD ID	SP FI	CT RS CI DD	SI KP DW
S22 Independent process audit and findings	IA	IA	PD	RS LC SP	DW KP
S23 Review proposed deferral, stop or exit	SI	PD SP	CI DD FI	RS LC CT	KP OG
S24 Company stop decision per articles	OG	SP	LC FI	RS CI DD SI	KP PD

Rows S11 and S12 apply only when regulations require those approvals. The roles of the President and the Supervisory Board are those set by regulation; they are shown for completeness and are not assigned by this study.

13.7 Blocker resolution and escalation



These are internal targets, not legal deadlines or permit promises. Delay never creates automatic approval. Reports separate review time from time spent waiting for sponsors, vendors, regulators or funders, and each escalation records the outcome, the reason and who must act [D2 §11].

13.8 The Deputy Head question

PP 19/2026 does not explicitly create a Deputy Head (Wakil Kepala) of the Executive Body. The Head takes Executive Body decisions; members are appointed by the President; Art. 14 allows members “as needed” for other portfolios [D2 §7].

Design	Instruments to review	Limit of authority
Strategic Industry unit with a professional leader	Head's regulation, unit charter, budget, board manual, staffing rules	Unit leader is not an organ member merely by holding a senior title
Executive Body member for Strategic Industry	Need under Art. 14; Presidential appointment; portfolio and work instruments	May lead coordination; membership does not transfer the Head's decisions
Formal Deputy Head	Harmonisation of Arts. 12–18 and related SOE Law provisions; PP amendment if the position changes those arrangements	A Presidential Decree, Regulation or internal rule cannot override higher law
New Strategic Industry holding	Presidential approval; purpose, articles, shares, capital and asset transfers as required (Art. 29B)	A holding does not automatically give an official authority over all projects

Nomenclature and escalation criteria

Appointments to the Executive Body are made by the President. Use “Head of Strategic Industry and Project Delivery” for a professional unit leader, or “Member of the Executive Body for Strategic Industries” if so appointed. If a formal position is later created, the clear English equivalent is “Deputy Head of the Executive Body for Strategic Industries and Project Delivery”. Avoid “Deputy Chairman”, which can be confused with the Deputy Chair of the Supervisory Board; an executive leader should not chair oversight of their own work [D2].

Review a formal position no earlier than month 12, and only when evidence shows: unresolved mandate disputes, recurring cross-arm decision queues, a need for a deputy to the Head, or a scope that genuinely requires a formal office. Measure decision times before and after to confirm the unit has not simply moved the queue.

13.9 Lessons from other countries

Country / institution	Practice	Adaptation for Danantara
China: SASAC, SDIC	Companies are the subject of investment decisions and accountability; investment plans, prohibited and specially supervised lists, feasibility studies, post-investment evaluation; SDIC builds industrial ecosystems through direct investment and funds	Clear decision ownership per entity; transparent entry criteria; value-chain platforms. Not transferable wholesale; Danantara still needs market tests and stop discipline
Vietnam: CMSC, SCIC	CMSC (2018, 19 enterprises) dissolved effective 21 March 2025 and enterprises returned to ministries; June 2026 direction towards a national investment fund based on SCIC	Any reorganisation needs a transfer list of authorities, project files, contracts, assets, staff and liabilities. CMSC's dissolution is not proof that central ownership fails
India: PRAGATI, NIIF	Technology-enabled monitoring with top-level leadership to clear blockers; NIIF as a government-anchored platform attracting institutional capital with due diligence before investment	Separate the blocker-resolution forum from investment testing; meetings do not replace permits or company decisions
Malaysia: Khazanah, NIMP 2030	Board charter, schedule of matters, limits of authority, risk framework; mandates differentiated (Investments Portfolio, Dana Impak, Developmental Assets, Special Situation Assets); Delivery Management Unit at MITI	Mandate contracts per arm; national impact targets alongside financial targets and risk limits

Source: D2 §4–5 and references R10–R19 therein.

13.10 Incentives

Tie variable pay to net value creation and verified operating and outcome improvements, with deferred assessment for long-cycle projects, risk and integrity conditions, adjustment for restatements, and recovery of awards where applicable. Avoid formulas based mainly on funds raised, MoUs signed, capex spent or dividends extracted [D1].

14
Performance management and measurement

Three linked scorecards, platform capability indicators, a fair rating system and controls against double counting.

14.1 Three linked scorecards

Scorecard	Metrics	Proposed weight	Examples
Investment	I01–I12	40%	Net return above benchmark; net IRR above hurdle; PME; DPI; financial NPV; liquidity coverage; attributable private mobilisation
SOE operations	O01–O15	35%	ROIC less WACC; incremental economic profit; cash conversion; real unit cost; bank ROE less cost of equity; unfunded PSO cost
Economic and industrial development	D01–D12	25%	Economic return above social hurdle; additional value added; sustained jobs; net FX; technology milestones; public support per outcome
Governance gates	6 controls	Pass / fail	Gates, not bonus points; an unresolved material gate prevents an unqualified overall score

Source: D1. Weights are a governance starting point only. Full metric dictionary in Appendix A.

D1's 40/35/25 weighting reflects where data will be available first. It does not reflect the priority of the industrial mandate. We recommend publishing a glide path now: move to about **30/30/40** by 2028, as development and capability data become rateable, and fold the platform capability indicators (Section 14.2) into the development stream when their evidence meets the eligibility rules.

Scoring rules

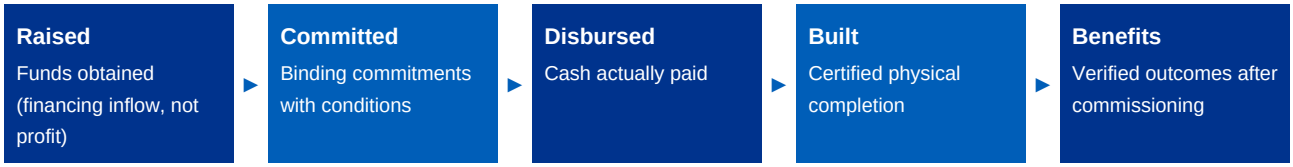
- Each metric has approved adverse, target and stretch anchors, scored **0, 80 and 100**, with linear interpolation capped at 0–100; lower-is-better metrics are reversed.
- At least **80% of applicable metric weight** must be rateable in each stream; all critical metrics and gates must pass before a composite score is published.
- **Missing metrics stay NR, never zero.** Exclusions need a reason and approval; weights may not be removed because a result is poor.
- Only class A or B evidence with comparable scope, an approved benchmark and a source is eligible for scoring.
- **No current aggregate score is assigned**: the system is ready for management data, but complete portfolio observations and approved benchmarks are not yet available [D1].

14.2 Platform capability indicators

D1's scorecards measure the three mandates. The platform model adds a fourth view: whether DOCK-UP is building suppliers. We propose five capability indicators, reported with the development scorecard:

Indicator	Definition	Evidence
Suppliers by gate	Number of SMEs at Entry, G1, G2, G3, G4, and movement per year	Independent assessor records in the shared register
Domestic value-added share in anchor packages	Verified domestic value added / package value, by anchor project (consistent with D05)	Localisation plan, invoices and value-added verification
Supplier productivity	Real value added per FTE of DOCK-UP suppliers versus a comparison group (consistent with D09)	Supplier financial data; cohort comparison
Cost parity	Supplier unit price / landed import price for the same specification	Purchase records; learning-curve tracking
Repeat and export orders	Value of orders outside the original anchor programme	Contracts and export documents

14.3 Five capital stages reported separately



Fundraising and investment spending are not profit. Each stage needs its own evidence, and a later stage can never be inferred from an earlier one [D1].

14.4 Controls against double counting

Potential misstatement	Required control
Asset transfers called investment returns	Bridge: opening NAV + external capital + investment result − distributions = closing NAV; explain FX and perimeter changes separately
SOE profit and its dividend both added as group income	Apply consolidation eliminations; report cash remittance as a separate financing-capacity measure
Entire project cost called Danantara investment	Record total capex, Danantara equity/debt, other public capital and private funding separately
Same investment claimed by multiple co-financiers	MDB-aligned attribution; distinguish direct mobilisation, indirect co-finance and total financing
Project sales, supplier sales and GDP benefits summed	Use value added and a reconciled input–output framework; deduct imports and intermediate transactions
Gross job counts across the supply chain	Separate direct FTE, temporary person-years and modelled indirect jobs; identify overlap and displacement
A national GDP change called Danantara impact	Specify without-project scenarios; report contribution ranges, not unsupported point estimates

Source: D1.

14.5 Fifteen leadership KPIs

For the Head of the Executive Body and the Supervisory Board, we propose fifteen KPIs that connect data, decisions, investment, SOEs, development, suppliers and financing. Targets should be set after baselines are established.

#	KPI	Source metric / definition
1	Register quality	Share of projects with confirmed sponsor, scope, stage, cost, funding and schedule (100% of identities mapped in 30 days) [D2]
2	Decision time	Median and 90th percentile from complete documents to decision; waiting time separate (proposed −30% median in six months, after baseline) [D2]
3	Blocker resolution	Issues lawfully resolved / issues due, with ageing
4	Evidence coverage	Share of applicable metric weight rateable, per stream (≥ 80%)
5	Investment return	I01 and I02: net return above benchmark; net IRR above hurdle
6	SOE value creation	O02: incremental economic profit
7	Construction SOE cash health	O04 operating cash conversion, days sales outstanding, interest coverage and liquidity runway
8	Delivery	D08: on-time, on-budget commissioning
9	Additional value added	D02: additional domestic value added, real
10	Supplier progression	DOCK-UP suppliers reaching G2 or above
11	Localisation	D05: domestic value-added share in anchor packages
12	Technology	D06: verified technology capability milestones
13	Jobs	D03: sustained additional employment
14	Resource-finance discipline	Coverage ratio of NRF instruments; present value of revenue given up, disclosed
15	Domestic financing share	Share of strategic-programme finance raised domestically versus new foreign loans (substitution test)

14.6 Reporting cadence

Frequency	Content
Monthly	Liquidity, construction progress, operating exceptions, critical risks
Quarterly	Investment returns, SOE transformation, project cost and schedule, evidence coverage, DOCK-UP gate movements
Annually	Audited financial statements, outcome evaluation, peer review, capital reallocation, public portfolio report
At commissioning and 12–24 months later	Technical performance, commercial ramp-up and benefits realisation

15
Key risks and mitigations

Risks that could defeat the transformation, with the controls proposed in this study.

Table 21: Risk register

Ratings are qualitative judgements of likelihood (L) and impact (I) over the next 24 months: High, Medium or Low. Owners are Danantara functions; where another institution must act, the Danantara owner coordinates with it.

#	Risk	How it would materialise	Mitigation	L / I	Owner
1	Quota instead of gates	Orders placed before verified readiness; quality failures on warships or infrastructure	Gate rules in contracts; independent assessors; pre-approved fall-back sources	M / H	Unit (with integrators)
2	Mandate duplication	Unit becomes an extra meeting layer; arms dispute ownership	Charters per function; single decision package; stop recurring meetings	M / M	Head of the Executive Body
3	Promoter dominance	Project champions override risk and legal objections	Independent risk and legal lines; dissent recorded in minutes	M / H	Risk; Legal
4	Hidden fiscal liabilities	NRF or project notes treated as non-debt; contingent claims surface later	Substance-based classification; pledged-revenue register; disclosure	M / H	Finance, with the Ministry of Finance
5	Legal challenge to resource rights	Licence holders or investors contest pledges or licence reviews	Pre-issuance legal review; due process; issue only on undisputed assets	M / H	Legal, with the Government
6	Overstated valuations	Headline reserve values read as financing capacity	KCMI/CRIRSCO proven and probable only; independent valuers; coverage caps	H / H	Finance (with an NRF board if created)
7	Programme delay or late state payment	A milestone or state payment slips; project-note investors unpaid	Liquidity lines; stress tests; conservative tranche sizing	M / M	DIM (with the issuer)
8	Technology and cyber	Smart-contract fault, key compromise, data leak	Independent audits; HSM key management; BSSN controls; fiat fall-back ledger	L / H	CTO (with the issuer)
9	Construction SOE contagion	Losses impair state-bank asset quality and SOE dividends	Group stress tests; legacy-asset separation; concentration limits	H / H	DAM; Risk
10	AI model and data risk	Agents act on poor master data or produce unsupported answers	Read-only start; source citations; evaluation sets; human approval	M / M	AI platform owner
11	Adoption failure	Dashboards built but not used for decisions	Business team in the build; decisions logged with agent evidence	M / M	DAM (with SOE CEOs)
12	Misuse of the strategic label	Weak projects enter the portfolio to gain support	Entry criteria; re-evaluation; stop and exit discipline	M / H	Unit; Supervisory Board

#	Risk	How it would materialise	Mitigation	L / I	Owner
13	Vendor and technology dependence	Foreign partners retain IP; local capability not built	IP strategy at P2/P3; independently accepted technology milestones	H / M	CTO
14	Claims without evidence	Announcements reported as results	Evidence classes; NR treatment; capital-stage reporting	H / M	Performance office
15	Sovereign rating and fiscal space	Rating agencies treat NRF notes or project notes as government debt	Early dialogue with the Ministry of Finance; conservative issuance; disclosure	M / H	Finance
16	Overlapping institutional mandates	NRF, Danantara and ministries contest roles	Mandate letters; owner–manager separation; agreed coordination forum	M / H	Head of the Executive Body
17	Leadership transition and key people	Programme loses its sponsor or scarce specialists	Documented mandates; succession plans; institutional rather than personal ownership	M / M	Head of the Executive Body; HR
18	Procurement and integrity in DOCK-UP	Gate results or orders influenced improperly	Independent assessors; rotation; audit trail in the register; conflict declarations	M / H	Internal Audit; unit
19	SME credit concentration	SME defaults cluster in banks lending against gates	Credit limits per programme; first-loss sizing; monitoring	M / M	Risk, with lending banks

Table 22: Risk map (risk numbers by likelihood and impact)

Likelihood \ Impact	Low impact	Medium impact	High impact
High likelihood	–	13, 14	6, 9
Medium likelihood	–	2, 7, 10, 11, 17, 19	1, 3, 4, 5, 12, 15, 16, 18
Low likelihood	–	–	8

16
Integrated roadmap

Eight workstreams in four phases over 24 months and beyond, with decision points that make every expansion conditional on evidence.

16.1 Workstreams and phases

Workstream	0–90 days	4–12 months	12–24 months	> 24 months
1 Governance and unit	Approval in principle (Day 0); charter; leader; secondments; authorisation map; pilot project group; design confirmed Day 90	Extend unit to full portfolio; assurance review; review of any formal position no earlier than month 12	Decide permanent structure on evidence	Review holding options only if criteria met
2 Register and scorecards	Reconciled register; capital/perimeter bridge; data dictionary; benchmark policy	First reconciled dashboard; quarterly scorecards	Published annual portfolio report	Completed-project evaluations
3 DOCK-UP	Gates and pilot design adopted (Day 60); assessor pool; entry assessments	Pilot starts month 4: 1 integrator, 20–30 SMEs, 3–5 component groups; first G1 orders around months 10–11	Decision on second integrator and civil anchors (months 12–15); first G2–G3 suppliers	Standard model across DEFEND ID and anchor projects
4 Anchor projects	Identify anchors; long-list localisable packages	Localisation plans at P2 for priority anchors	Framework contracts at P3/P4	Verified value-added reporting at P6
5 NRF	Proposal for the Government's consideration (Day 90)	If the Government proceeds: Stage 0 (rights, reserves, valuation, fiscal treatment), then a Stage 1 pilot if the gate is met	Stage 2: strategic minerals; industrial and DOCK-UP windows	Stage 3: mature fund
6 Project tokenisation	Work on six pre-conditions; corrected term sheet	Phase 1 pilot (US$150 m) in the OJK sandbox once pre-conditions are certified (earliest Q1 2027); compare with conventional escrow	Phase 2 decision on interim evidence; IDR receivable notes for other sectors	Phase 3 distribution only after licence and pilot maturity
7 Construction SOEs	Audited diagnostic (Day 30–90); bid discipline rules	Legacy-asset separation; cash protection	Structure decision on evidence	Sustained economic profit
8 AI platform	MVP starts Day 30; 6 agents; pilot on 2–3 projects live about Day 120	Phase 2 (6–12 months after MVP start): 2–3 SOEs; 20–50 projects	Phase 3: all construction SOEs; 100+ projects	Phase 4: national platform; extend to DOCK-UP, NRF, scorecards

16.2 Decision points

When	Decision	Evidence required
Day 90	Confirm, adjust or stop the unit design	Decision-time data; resolved and unresolved blockers; decision packages with risk and legal opinions
About month 6 (after MVP evaluation)	Scale AI to Phase 2	MVP outcomes; net benefits measured
Month 12 or later	Review whether a formal Executive Body position is needed	Criteria in Section 13.8, measured over at least two quarters
Months 12–15	Extend DOCK-UP to a second integrator and to civil anchors	Pilot suppliers' gate results; quality records; integrator feedback
When Stage 0 is complete	Proceed from NRF Stage 0 to a pilot instrument (Government decision)	Legal rights confirmed; independent valuation; fiscal classification
Interim, during the pilot	PALCapital Phase 2	Release controls and cost comparison proven; full credit performance known only at the pilot's 720-day maturity

The previously discussed planning direction (institutional preparation in 2026, development of financing in 2027 and industrialisation from 2028) can be retained as a guide. Projects that are ready need not wait for the next calendar stage, and resource-based financing should open only after the review of rights, revenue, risk and authority is complete [D2].

16.3 First 90 days in detail

Period	Deliverable	Accountable / acceptance
Days 1–45	Draft (Day 30) and approved (Day 45) ownership and project register; capital and perimeter bridge; data dictionary; three mandate statements; unit charter; pilot project group selected	CFO as interim owner with DAM, DIM and DDMF, handing over to the MD Portfolio and Project Delivery once appointed
Days 1–30	Investor-return methodology and benchmark policy; critical risk and assurance gates	Investment and risk committees
Days 31–60	Populate high-exposure SOEs and all active strategic projects; reconcile cash, commitments and physical milestones; parallel reviews, dashboard and blocker list	Arm heads and project CFOs: source-backed actuals and explicit NR entries
Days 31–60	Independent appraisal of the largest fiscal and technology risks; public-support and additionality cases; NRF proposal for the Government prepared	DDMF economists and Risk; Legal
Days 61–90	First reconciled dashboard; independent assurance review; evaluation of pilot projects; construction SOE diagnostic report; AI MVP in build (pilot live about Day 120)	CFO with Audit: coverage and qualification statement signed; unit evaluation report

17
Recommendations by stakeholder

What each stakeholder is asked to do.

Stakeholder	Recommended actions
For the Government's consideration	Consider the National Industrial Capability Platform as Danantara's industrialisation model. Consider a legal, valuation and fiscal review before establishing an NRF, with separate owner and manager roles. Consider domestic-financing substitution tests before new foreign loans for strategic programmes.
For the Ministry of Finance's consideration	Consider classifying resource-backed and receivable-backed instruments by substance in the fiscal-risk statement. Consider defining a consent process for assigning state contract receivables to collection accounts.
Supervisory Board	Approve the three-scorecard framework, NR treatment and governance gates. Require an audited opening-balance bridge and a published annual portfolio report. Oversee the 90-day unit test and the evidence for any Deputy Head proposal.
Head of the Executive Body	Establish the Strategic Industry and Project Delivery unit (implementing NITO) with a limited charter. Approve the reconciled 40-project register as the single denominator. Chair cross-arm escalations; keep risk, legal and audit independent.
DAM / COO	Launch the construction SOE diagnostic and legacy-asset separation. Sponsor the AI MVP and Phase 2 rollout.
DIM / CIO	Publish return methodology by capital pool; apply investment discipline to project notes and NRF windows. Run the PALCapital pilot only after the six pre-conditions are met.
DDMF / CTO	Require financial and economic cases with explicit public support for development projects. Provide independent technical review and own technology-milestone acceptance with an independent panel.
For the Ministry of Defence and KKIP's consideration	Consider DOCK-UP as the standard SME participation model, with gate-based packages in framework and offset contracts. Consider supporting assignment of contract receivables to milestone escrow.
For OJK and Bank Indonesia's consideration	Consider the PALCapital pilot under the IKD sandbox with regulator nodes. Consider the treatment of tokenised receivable notes for institutional investors.
Integrators (PT PAL, DEFEND ID members)	Publish localisation plans; place orders only on verified gates. Open supplier portals linked to ERP; share drawings and gate records.
Construction SOEs	Adopt bid discipline and cash protection rules; implement the cost-code backbone and AI pilot.
Banks, LPEI and guarantee providers	Link KUR, supply-chain finance, investment credit and export finance to DOCK-UP gates; co-design first-loss guarantees.
SMEs	Enter the readiness register; follow the development plan to G1–G4; invest in process quality first.

Stakeholder	Recommended actions
BKI, test laboratories, universities and BRIN	Form the independent assessor pool; support TRL maturation and certification.

Closing

Danantara does not need more announcements to prove its relevance. It needs a mechanism that converts capital into capability, financing that respects the substance of what is borrowed, and evidence that the results are real. The proposals in this study are designed to be tested quickly, measured honestly, and expanded only where they work.

A
Appendix A: Metric dictionary

The 39 result metrics proposed in D1, in three scorecards. All thresholds require a defined reporting scope and approval; a peer's return or a generic sector ratio is not automatically a Danantara hurdle.

A.1 Investment metrics

Code	Metric (unit)	Measurement	Benchmark and caution
I01	Net public-market return above policy benchmark (pp)	Geometrically linked TWR net of all investment costs minus matched policy-index return	Same dates, currency, total-return basis and mix; since inception and rolling 3/5 years; no annualised short-history claims
I02	Private investment net IRR above hurdle (pp)	XIRR of dated investor cash flows incl. independently reviewed residual NAV, less approved equity hurdle	Vintage, sector, geography and leverage matched; do not average IRRs; pool cash flows
I03	Private investment PME (×)	Kaplan–Schoar PME using index-adjusted contributions, distributions and NAV	PME > 1 against a preselected total-return index; PME is not an IRR
I04	Distributions to paid-in capital (×)	Cumulative distributions / paid-in capital; show TVPI and RVPI	Vintage-specific; early greenfield DPI can be zero without failure
I05	Financial NPV / invested capital (%)	Full-life NPV of project FCF at approved WACC / PV of investment outlays	Separate forecast appraisal from ex-post results; exclude sunk costs for incremental decisions
I06	Credit and investment loss ratio (%)	Credit impairment net of reversals and recoveries / average gross exposure	Matched rating, tenor and product; equity write-downs separate
I07	Exposure above concentration limits (%)	Exposure exceeding approved single-name/sector limits / portfolio NAV	Zero unauthorised excess; stress related-party and correlated exposure
I08	Stressed liquidity coverage (×)	Unencumbered cash + stressed liquid assets + usable committed lines / stressed 12-month needs	Board-approved floor; exclude restricted cash and conditional facilities
I09	Total investment cost ratio (bps)	Management, external manager, transaction and support costs / average NAV × 10,000	Same asset class and model; low cost alone does not prove better net performance
I10	Attributable private mobilisation (×)	Eligible private direct mobilisation attributable to Danantara / own committed capital	MDB-aligned attribution; MoUs and total project value do not count
I11	Strategic investment milestone delivery (%)	Capital-weighted milestones independently accepted / milestones due	Frozen baseline; signing or inauguration does not certify financial close or progress
I12	Funding spread over matched sovereign curve (bps)	Issue-yield spread against same-currency, duration-matched sovereign curve	A funding-access metric; cannot substitute for asset returns or WACC

A.2 SOE metrics

Code	Metric (unit)	Measurement	Benchmark and caution
O01	Non-financial SOE ROIC less WACC (pp)	Normalised after-tax operating profit / average invested capital − WACC	Sector- and currency-specific; exclude banks
O02	Incremental economic profit (Rp bn)	NOPAT − WACC × average invested capital, minus counterfactual economic profit	Ownership transfers and capital injections are not value creation
O03	Comparable EBITDA margin (%)	Comparable EBITDA / revenue, bridged to statutory operating profit	Show reported and normalised; exclude banks
O04	Operating cash conversion (×)	Operating cash flow / EBITDA, with tax, working-capital and maintenance-capex bridge	Do not reward overdue suppliers or uncollected compensation
O05	Real unit operating cost (index)	Controllable costs at constant prices / quality-adjusted output; baseline 100	Savings must not come from deferred maintenance, safety or quality erosion
O06	Real value added per employee (Rp m/FTE)	Real gross value added / average FTE	Revenue per employee can be inflated by prices or outsourcing
O07	Organic real revenue growth (%)	Like-for-like constant-price growth excluding M&A and accounting changes	Separate price, volume, FX, consolidation and subsidy effects
O08	Non-financial net debt / EBITDA (×)	Interest-bearing debt less unrestricted cash / trailing comparable EBITDA	Not meaningful for banks or negative EBITDA
O09	Verified net transformation benefits (Rp bn)	Realised cash benefits less implementation, severance, integration and recurring costs	Entity-count reduction is not a benefit
O10	Service reliability / quality attainment (%)	Accepted sector service outcomes / contracted targets	A simple average must not conceal critical failures
O11	Bank ROE less cost of equity (pp)	Profit to common equity / average common equity − cost of equity	Do not raise leverage merely to raise ROE
O12	Bank cost-to-income ratio (%)	Operating expense / operating income on a consistent definition	Distinguish CIR from BOPO
O13	Bank gross NPL ratio (%)	Gross NPL / gross loans; show cost of risk and coverage	Loan sales, write-offs and rapid growth can mask deterioration
O14	Bank capital headroom (pp)	CET1 ratio minus applicable required CET1 including buffers	Total CAR and Tier 1 are not interchangeable with CET1
O15	Unfunded public-service mandate cost (Rp bn)	Efficient verified PSO cost less funded compensation; arrears aged	Subsidies are not efficiency gains

A.3 Economic and industrial metrics

Code	Metric (unit)	Measurement	Benchmark and caution
D01	Economic return above social hurdle (pp)	EIRR of incremental economic flows minus approved social discount rate; disclose ENPV	ADB 2017: 9% reference, 6% exception; requires local policy approval
D02	Additional domestic value added (Rp bn real)	Value added with project minus credible without-project path, net of displacement	Gross sales and investment spending are not GDP increments
D03	Sustained additional employment (FTE)	Ongoing direct FTE above counterfactual, net of displacement; construction person-years separate	Avoid counting a worker in contractor, project and supplier totals

Code	Metric (unit)	Measurement	Benchmark and caution
D04	Net foreign-exchange contribution (US$ m)	Exports + avoided imports − imported inputs/capex − external fees and remittances	Present current-account contribution and financing flows separately
D05	Domestic value-added share (%)	Verified domestic value added / production value	Not identical to TKDN certification or local invoice value
D06	Verified technology capability milestones (%)	Weighted milestones accepted / due: design rights, process control, certified local supply, repeat sales	Training, patents and MoUs alone do not show mastery
D07	Commercial capacity utilisation (%)	Saleable accepted output / practical capacity after commissioning	No output credit before commercial operation
D08	On-time / on-budget commissioning (%)	Baseline-capital-weighted capacity meeting original cost and date / capacity due	Rebaselining cannot erase overruns
D09	Supplier productivity improvement (%)	Matched suppliers' real value added per FTE change less comparison-group change	Selection effects can mimic improvement
D10	Net lifecycle GHG reduction (tCO2e/yr)	Baseline lifecycle emissions minus project lifecycle emissions	Designed avoidance is not realised abatement
D11	Reliable affordable service delivered (units)	Incremental users or units meeting quality, affordability and access thresholds	Connections or installed assets alone are insufficient
D12	Public support per additional outcome (Rp/outcome)	PV of grants, concessionality and expected guarantee cost / incremental outcome	Least-cost alternative; one outcome denominator

A.4 Governance gates (proposed)

D1 specifies six governance controls that act as gates but does not name them in the material reviewed. We propose the following six, to be confirmed:

1. Approved and reconciled project and ownership register.
2. Audited financial statements and opening-balance bridge for the reporting perimeter.
3. Independent valuation of material private and strategic holdings.
4. Conflict-of-interest register with recusal records for all material decisions.
5. Independent risk and legal opinions recorded for all decisions above threshold.
6. Data-assurance statement on evidence coverage and qualifications.

B

Appendix B: Discrepancies in the source documents

Inconsistencies found during this review. None has been resolved by assumption; each must be corrected by its owner before the figure is used for decisions.

#	Source	Discrepancy	Action
1	D2 / U1	40-project totals: summary Rp2,091.46 tn versus detail Rp2,072.53 tn (−Rp18.93 tn); cluster allocations differ (Others +Rp291.44 tn; Minerals −Rp196.98 tn)	Data owner to confirm units, scope and allocation
2	D5	2026 natural-resource share: chart label 10.86% versus text 10.68%	Correct one value
3	D5	Rp100 tn / Rp1,000 tn described as “direct fiscal benefit” in the text, but shown in the chart under “coal capitalisation value with NRF” beside “Coal Reserve Capitalization” Rp10,000 tn	Separate stock (capitalisation) from flow (fiscal benefit)
4	D5	“Minimum valuation Rp10,000 tn” has no stated method or reserve category	Publish method under KCMI/CRIRSCO
5	D6	“Not a grant, not debt” versus “yield-bearing debt note” and “secured creditor”	Classify by substance as debt
6	D6	“No collateral” versus “MoD PO collateral”	Clarify the security package
7	D6	“6–7% fixed yield” versus “700 bps”; unexplained “+200 bps”	Corrected term sheet
8	D6	Go-live “45 days” versus “90 days”	Single plan
9	D6	ZK proof of “US$500 m value” versus a US$150 m issue	Explain the relationship (contract value versus issue size)
10	D6	Danantara “US$900 bn+ AUM”, “US$20 bn initial”, “1,000+ SOEs”; “pre-cleared by BSSN and OJK”	Provide sources and written confirmations
11	D3 p.7 / p.8	Technical team of 14: p.7 includes a project lead and combines DevOps and security; p.8 has no project lead and separates security/IAM	Confirm roles
12	D3 p.7	Gantt chart shows M1–M12 inside a three-month plan	Label as weeks
13	D3 p.5 / p.7	Six target agents (portfolio risk, bid/no-bid, cost estimating, receivables, schedule, procurement) differ from the six MVP agents (project document, cost, cash and receivables, procurement, schedule, PM orchestrator)	Confirm phasing
14	D3 p.8	Llama 3.1 70B (4-bit) listed under 30–40B models	Load-test or reclassify
15	D3 p.3	“Industrial estates” listed twice among burdensome assets	Editorial
16	D3 p.3	Infrastructure budget fall shown as −45%; computed −44.7%	Rounding; acceptable
17	D4 p.7	“Circle = month” label while the TRL route uses step numbers	Clarify the timeline of the TRL route

#	Source	Discrepancy	Action
18	D3 p.8	Phase 1 sized at “4 GPU (1 site)”, but the specification table calls for 1–2 servers of 4 × L40S each (4–8 GPUs)	Confirm Phase-1 GPU count
19	D5	Revenue-share series presented without a stated source	Reconcile to budget data before use
20	D3 p.7–8	Business team stated as 6–8 people, but the roles listed add up to 7–8	Confirm team size

C
Appendix C: Calculations

Calculations performed for this study. Inputs are from the sources unless marked as illustrative (class D).

Item	Calculation	Result
DIM notes weighted yield	(750 × 5.35% + 750 × 5.95%) / 1,500	5.65%
DIM notes average spread	(32 + 34) / 2	33 bps
DIM notes cover	4.6 / 1.5	≈ 3.07×
CA–EDC joint share	200 / 800	25%
40-project difference	2,091.46 − 2,072.53	Rp18.93 tn
Construction SOE losses	10.14 + 8.0 + 5.4 + 4.48	Rp28.02 tn
Infrastructure budget change	(61.38 − 110.95) / 110.95	−44.7%
NRF period averages	Mean of 2004–2014 annual shares; mean of 2015–2024	19.3%; 7.7% (reconcile with D5)
Implied 2026 state revenue	336 / 10.68%	≈ Rp3,146 tn
Royalty versus potential (D5's comparison; a flow against a stock)	126.4 / 10,000	1.26% (D5: “< 2%”); not a measure of capture
NRF illustration (class D)	50 Mt × US$80 × 15% = US$0.6 bn/yr; × 20-yr annuity factor at 8% (9.818) = US$5.89 bn; ÷ 1.5	US$3.93 bn; 4.9% of US$80 bn
PAL-SUB-1 yield	150 × 7% × 720 / 365	US$20.7 m
PAL waterfall example	5.17 / (150 × 7%) × 365	≈ 180 days of accrual
PAL savings check	150 × (10.5% − 7%) × 720 / 365	≈ US$10.4 m (within the claimed US$8–12 m)
PAL tranche total	40 + 45 + 40 + 25	US$150 m
Learning curve (class D)	Unit cost = 0.85 to the power n; parity when 1.4 × 0.85 to the power n = 1	n ≈ 2.07 doublings ≈ 4.2× first-lot volume

Item	Calculation	Result
Sources and uses (class D)	7 × 5 × 16,500 = Rp578 tn; 20% × 126.4 = 25.3 × annuity(10y, 7%) 7.024 = 178 ÷ 1.5 = 118; 1.45 bn × 16,500 = 24	New channels Rp142 tn (6.8%); remaining Rp1,371 tn
Equity and debt split (class D)	30% × 2,091.46; 70% × 2,091.46	Rp627 tn equity; Rp1,464 tn debt
Localisation scenario (class D)	100 × 45% × 60% = 27 imported; × 10/20/30% shifted; × 40% value added; × 0.7 net FX; ÷ Rp0.25 bn per job-year	Per Rp100 tn: VA 1.1 / 2.2 / 3.2 tn
AI receivables illustration (class D)	20,000 × 10 / 365; × 10%	Rp0.55 tn released; Rp55 bn a year
Financial vs economic example (D1)	−100 + 15 × annuity(10y, 12%) ; −100 + 21 × annuity(10y, 9%)	FNPV −Rp15.25 bn; ENPV Rp34.77 bn

D
Appendix D: Glossary

Additionality: whether an outcome would have happened without Danantara. Counterfactual: the credible without-project path used to measure additionality. Denominator: the agreed population (for example, the reconciled project register) against which rates are calculated. Perimeter bridge: a reconciliation explaining changes in what is included in the portfolio. Waterfall: the order in which incoming payments are applied. Coverage ratio: pledged cash flow divided by debt service. Advance rate: the share of pledged value that can be borrowed.

Term	Meaning
ADB / IFC / MDB	Asian Development Bank / International Finance Corporation / multilateral development bank
APBN	State budget
AUM / NAV	Assets under management / net asset value
BIM / CDE	Building information modelling / common data environment
BKI	Biro Klasifikasi Indonesia, the Indonesian classification society
BP BUMN	The SOE regulatory agency
BPI	Badan Pengelola Investasi Daya Anagata Nusantara (Danantara); its Executive Body is the Badan Pelaksana
BPS	Statistics Indonesia
BRIN	National Research and Innovation Agency
BSSN	National Cyber and Crypto Agency
BUMN Karya	State-owned construction companies
CA–EDC	Chlor-alkali and ethylene dichloride chemicals project

Term	Meaning
CEO / CFO / COO / CIO / CTO	Chief executive / financial / operating / investment / technology officer
CET1 / CAR	Common equity tier 1 / capital adequacy ratio (banks)
CIR / BOPO	Cost-to-income ratio / operating expense to operating income (Indonesian bank ratio)
CMSC / SCIC	Vietnam's former Commission for Management of State Capital / State Capital Investment Corporation
CPI / SPI / EAC / BAC / EV / AC	Cost and schedule performance indices; estimate and budget at completion; earned value; actual cost
CRIRSCO / KCMI	International reserves reporting template and the Indonesian code for reporting mineral resources and reserves
DAM / DIM / DDMF	Danantara Asset Management / Investment Management / Development Management Fund
DEFEND ID	Indonesia's defence-industry SOE holding
DOCK-UP	Develop, Off-taker, Certification, Upgrading supplier-development framework
DPI / TVPI / RVPI	Distributions, total value and residual value to paid-in capital
EBITDA	Earnings before interest, tax, depreciation and amortisation
EIRR / ENPV	Economic internal rate of return / economic net present value
EMC / IRST	Electromagnetic compatibility / infrared search and track
EPC	Engineering, procurement and construction
ERC-3643 / IBFT / FIPS	Permissioned security-token standard / Istanbul Byzantine fault-tolerant consensus / US federal cryptographic module standard
ERP	Enterprise resource planning system
FEED / FID	Front-end engineering design / final investment decision
FTE	Full-time equivalent
FX	Foreign exchange
GHG / tCO2e	Greenhouse gas / tonnes of carbon-dioxide equivalent
GMS	General meeting of shareholders (RUPS)
GPU / LLM / RAG / OCR	Graphics processor / large language model / retrieval-augmented generation / optical character recognition
HSE	Health, safety and environment
HSM / IAM	Hardware security module / identity and access management
IKD	OJK's digital financial innovation framework and sandbox
IMF / IFSWF / GIPS	International Monetary Fund / International Forum of Sovereign Wealth Funds / Global Investment Performance Standards
INA	Indonesia Investment Authority
IoT / SCADA	Internet of things / supervisory control and data acquisition
IP	Intellectual property
IRR / XIRR / TWR	Internal rate of return / IRR on dated cash flows / time-weighted return
IUP / HGU	Mining business licence / land-use right

Term	Meaning
KKIP	Defence Industry Policy Committee
KUR	Kredit Usaha Rakyat, subsidised people's business credit
KYB / AML	Know your business / anti-money laundering
LPD	Landing platform dock (ship)
LPEI	Indonesia Eximbank
MD	Managing Director
MIND ID	Indonesia's mining SOE holding
MoD / MoF	Ministry of Defence / Ministry of Finance
MoU	Memorandum of understanding
MRL / TRL (TKT)	Manufacturing / Technology Readiness Level (TKT is the Indonesian TRL)
MRO	Maintenance, repair and overhaul
MSME / SME (UMKM / UKM)	Micro, small and medium enterprises / small and medium enterprises
MVP / UAT / SOP	Minimum viable product / user acceptance testing / standard operating procedure
NIIF / PRAGATI	India's National Investment and Infrastructure Fund / project-monitoring platform
NIMP / MITI	Malaysia's New Industrial Master Plan 2030 / Ministry of Investment, Trade and Industry
NITO	Danantara National Industrial Transformation Office (proposed)
NOPAT / ROIC / WACC	Net operating profit after tax / return on invested capital / weighted average cost of capital
NPL	Non-performing loan
NPV	Net present value
NR	Not rateable
NRF	Natural Resource Fund
OJK	Financial Services Authority
PBS / WBS	Product breakdown structure / work breakdown structure
Pinjaman luar negeri	Foreign loan
PKPU	Court-supervised debt restructuring (suspension of debt payment obligations)
PLM	Product lifecycle management
PLN	State electricity company
PME	Public-market equivalent
PMN	State capital injection
PNBP	Non-tax state revenue
PP	Peraturan Pemerintah, government regulation
PPA / PPP	Power-purchase agreement / public–private partnership
PSEL / WtE	Waste-to-electricity plant / waste-to-energy
PSO	Public-service obligation

Term	Meaning
RASCI	Responsible, Accountable, Supporting, Consulted, Informed
ROE	Return on equity
SASAC / SDIC	China's state asset regulator / State Development and Investment Corporation
SBN	Surat Berharga Negara, government securities
SPV	Special-purpose vehicle
TKDN	Domestic component level (local-content certification)
TNI AL	Indonesian Navy
ZK proof	Zero-knowledge proof: proves a fact without revealing the underlying data

E
Appendix E: References

Underlying references cited by the source documents and relied on in this study.

D3 to D6 cite no external references of their own, so only D1 and D2 references are listed. Law 1/2025 (Section 3.1) is cited from public law and should be verified by Danantara Legal.

E.1 Cited through D1 (performance study)

Code	Reference
S01	Danantara Indonesia: official website and reports landing page (accessed 6 Oct 2026)
S02	Danantara Indonesia: SOEs complete FY2025 statements; consolidation continues (2 Jul 2026)
S03	Danantara Indonesia: 2026 work plan and budget (1 Dec 2025)
S04	Danantara Indonesia: governance reset and strengthening SOE fundamentals (5 Mar 2026)
S05	Danantara Indonesia: US$1.5 billion international bond issuance (12 Jun 2026)
S06	Danantara Indonesia: Danantara and INA sign CA–EDC investment agreement (3 Mar 2026)
S07	Danantara Indonesia: construction of PSEL Bogor Raya 1 begins (17 Sep 2026)
S08	MIND ID: six Phase-I downstream projects, up to US$7 billion (event 6 Feb 2026)
S09	Danantara Indonesia: DDMF mandate and approach
S10–S13	BRI, Bank Mandiri, Telkom and PLN financial disclosures (2025–2026)
S14	BPS: Indonesia economic growth in 2025 (5 Feb 2026)
S15–S16	CFA Institute GIPS Standards for Asset Owners; IFSWF Santiago Principles
S17–S19	Temasek investment approach and Temasek Review 2026; Khazanah Annual Review 2026

Code	Reference
S20–S23	OECD SOE Guidelines 2024; IMF SOE Health Check Tool and fiscal-risk guidance
S24–S29	IFC AIMM; ADB Guidelines for the Economic Analysis of Projects (2017); IDB Invest DELTA; MDB mobilisation reference guide; UNIDO industrial indicators
S30–S31	Danantara Indonesia: Batch II WtE conditional awards (21 Jul 2026); investor-relations directory

E.2 Cited through D2 (governance evaluation)

Code	Reference
R1–R3	PP 10/2025; PP 19/2026; Law 16/2025 amending the SOE Law
R4–R8	Danantara structures of BPI, DAM, DIM and DDMF; official statement of 12 May 2026
R9	OECD Guidelines on Corporate Governance of State-Owned Enterprises 2024
R10–R11	SASAC measures for central enterprise investment; SDIC technology transformation fund
R12–R13	Vietnam: dissolution of CMSC; Ministry of Finance on SCIC restructuring (18 Jun 2026)
R14–R16	India: PRAGATI 50th meeting (31 Dec 2025); NIIF; AIIB NIIF PMF II appraisal
R17–R19	Khazanah governance framework and investment policy; Malaysia NIMP 2030 Delivery Management Unit